

Note

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1 February 26

1.1 Sample Preparation [1]

1.1.1 For $W_L^\pm W_L^\pm$

```
generate p p > w+{0} w+{0} j j QCD=0, w+ > l+ vl
add process p p > w-{0} w-{0} j j QCD=0, w- > l- vl~
output WW_LL
launch
shower = pythia8
detector = Delphes (R=0.4)
```

1.1.2 For $W_L^\pm W_T^\pm$

```
generate p p > w+{0} w+{T} j j QCD=0, w+ > l+ vl
add process p p > w-{0} w-{T} j j QCD=0, w- > l- vl~
output WW_LT
launch
shower = pythia8
detector = Delphes (R=0.4)
```

1.1.3 For $W_T^\pm W_T^\pm$

```
generate p p > w+{T} w+{T} j j QCD=0, w+ > l+ vl
add process p p > w-{T} w-{T} j j QCD=0, w- > l- vl~
output WW_TT
launch
shower = pythia8
detector = Delphes (R=0.4)
```

1.1.4 For Unpolarized $W^\pm W^\pm$

```
generate p p > w+ w+ j j QCD=0, w+ > l+ vl
add process p p > w- w- j j QCD=0, w- > l- vl~
output WW_UN
launch
shower = pythia8
detector = Delphes (R=0.4)
```

1.2 Restrictions for Candidates

1.2.1 Electrons

Electrons are required to satisfy the “Tight” likelihood-based identification criteria.

- $p_T > 27$ GeV
- $|\eta| < 2.47$ GeV
- Excluding the calorimeter transition region $1.37 < |\eta| < 1.52$

```
ele_mask = (eles.pt > 27) & (abs(eles.eta) < 2.47) & ~((abs(eles.eta) > 1.37) &  
(abs(eles.eta) < 1.52))
```

1.2.2 Muons

Muons are required to satisfy the “medium” identification selection.

- $p_T > 27$ GeV
- $|\eta| < 2.5$

```
mu_mask = (muons.pt > 27) & (abs(muons.eta) < 2.5)
```

1.2.3 Jets

Jets are reconstructed using the anti- k_t algorithm with a radius parameter of $R = 0.4$, using the particle-flow object.

- At least two jets with $|\eta| < 4.5$ are required to be present in each event
- $p_T > 65$ GeV for leading jets (written in the paragraph for election of the leading leptons later.)
- $p_T > 35$ GeV for subleading jets

```
jet_mask = (jets.pt > 35) & (abs(jets.eta) < 4.5)
```

1.3 Selection Cuts

The event selection in the signal region (SR) requires:

1. A same-sign lepton pair with $m_{\ell\ell} > 20$ GeV

```
padded_leps = ak.pad_none(all_leps, 2)  
l1, l2 = padded_leps[:, 0], padded_leps[:, 1]
```

```
mask1 = (ak.num(all_leps) >= 2) & ak.fill_none((l1.q == l2.q) & ((l1 + l2).mass  
> 20), False)  
n1 = ak.sum(mask1)
```

2. $E_T^{\text{miss}} > 30$ GeV

```
met_pt = ak.firsts(data.MissingET.MET)
mask2 = mask1 & ak.fill_none((met_pt > 30), False)
n2 = ak.sum(mask2)
```

3. VBS topology: The two highest- p_T jets must satisfy $m_{jj} > 500$ GeV with $|\Delta y_{jj}| > 2.0$

```
padded_jets = ak.pad_none(sel_jets, 2)
j1, j2 = padded_jets[:, 0], padded_jets[:, 1]

vbs_kinematics = (j1.pt > 65) & (j2.pt > 35) & ((j1 + j2).mass > 500) & (abs(j1.
rapidity - j2.rapidity) > 2.0)
mask3 = mask2 & (ak.num(sel_jets) >= 2) & ak.fill_none(vbs_kinematics, False)
n3 = ak.sum(mask3)
```

Cut	LL	pass rate	LT	pass rate	TT	pass rate	Unpol.	pass rate
Total	50000	1.00	50000	1.00	50000	1.00	50000	1.00
Lepton cut	12534	0.25	12453	0.25	12814	0.26	12491	0.25
MET cut	10327	0.21	10614	0.21	11156	0.22	10745	0.21
Jet cut	4403	0.09	5254	0.11	5477	0.11	5252	0.11

Table 1: Event Pass Rates across different polarization states.

1.4 Kinematic Distributions

1.4.1 Definition and Physical Meaning

1. Transverse Mass m_T

In the lab frame, we only know the transverse components of the neutrinos (the Missing Transverse Energy, or MET). Because the longitudinal momentum (p_z) of the neutrinos is unknown, we cannot calculate a true invariant mass. Instead, we project everything onto the 2D transverse plane ($x - y$) to calculate the Cluster Transverse Mass (m_T).

The Mathematics: To calculate the m_T of the WW system, we treat the two visible leptons as a single combined “cluster” ($\ell\ell$) and the MET as the invisible component. Dilepton Transverse Energy ($E_T^{\ell\ell}$): This combines the transverse momentum and the invariant mass of the dilepton system.

$$E_T^{\ell\ell} = \sqrt{|\vec{p}_T^{\ell\ell}|^2 + M_{\ell\ell}^2}, \quad (1)$$

Missing Transverse Energy (E_T^{miss}): This is simply the magnitude of the MET vector.

$$E_T^{\text{miss}} = |\vec{p}_T^{\text{miss}}|, \quad (2)$$

WW Transverse Mass (m_T^{WW}): We subtract the squared vector sum of the transverse momenta from the squared scalar sum of the transverse energies.

$$m_T^{WW} = \sqrt{(E_T^{\ell\ell} + E_T^{\text{miss}})^2 - |\vec{p}_T^{\ell\ell} + \vec{p}_T^{\text{miss}}|^2}. \quad (3)$$

2. Difference in Azimuthal Angle $\Delta\phi_{jj}$

$$\phi = \tan^{-1} \left(\frac{p_y}{p_x} \right) \rightarrow \Delta\phi_{jj} = |\phi_{j1} - \phi_{j2}|, \quad (4)$$

Measures the angle between the two tagging jets in the transverse plane.

3. Difference in Pseudorapidity $\Delta\eta_{\ell\ell}$

$$\eta = \frac{1}{2} \ln \left(\frac{|p| + p_z}{|p| - p_z} \right) \rightarrow \Delta\eta_{\ell\ell} = |\eta_{\ell1} - \eta_{\ell2}|, \quad (5)$$

Measures how far apart the two charged leptons are along the longitudinal axis.

1.4.2 Results

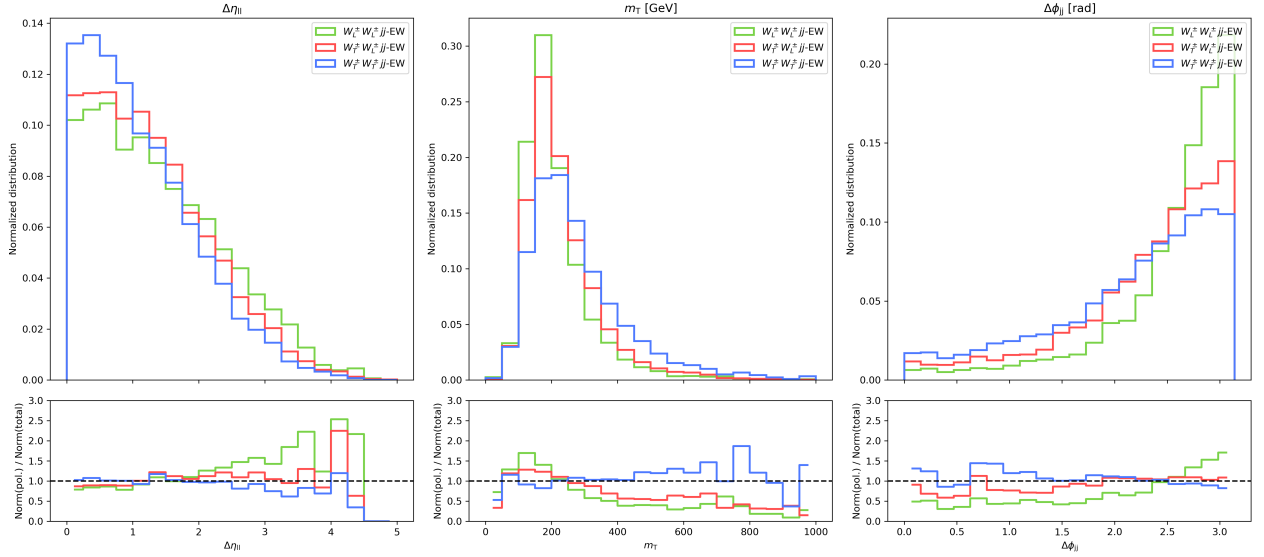


Figure 1: Distribution in the signal region for the difference in pseudorapidity between leading and subleading leptons, the transverse mass, and the difference in azimuthal angle between leading and subleading jets.

For the bottom panel, the ratio is calculated by comparing the normalized distribution of a polarized sample(LL, LT, or TT) against the normalized distribution of the unpolarized (total) sample. The ratio is derived for the plotting:

Before calculating the ratio, the code ensures both the polarized and unpolarized samples are on the same scale by normalizing them. Instead of using the raw number of events, it treats each event as a fraction of the total sample size:

$$w_i = \frac{1}{N_{\text{events}}}, \quad (6)$$

where N_{events} is the number of events that passed all cuts for that specific sample.

For each variable, the data is divided into k bins. Let $H_p(b)$ be the normalized count for a polarized sample in bin b , and $H_u(b)$ be the normalized count for the unpolarized sample:

$$H(b) = \sum_{x_i \in \text{bin } b} w_i, \quad (7)$$

The ratio plotted in the bottom panel represents how much a specific polarization deviates from the average unpolarized behavior in a specific kinematic region:

$$R(b) = \frac{\text{Norm(Polarized)}_b}{\text{Norm(Unpolarized)}_b} = \frac{H_p(b)}{H_u(b)}, \quad (8)$$

- If $R(b) > 1$: The polarized state is more likely to produce events in that specific parameter range than the total population.
- If $R(b) < 1$: The polarized state is suppressed in that range.

2 March 1st

2.1 Additional

To address the low selection efficiency observed in the initial MadGraph results¹, I implemented additional phase-space cuts directly in the run card and increased the number of events generated. This refinement ensures a higher yield of relevant events and optimizes computational resources.

I adjusted the following parameters in the run_card. Including restrictions for the transverse momentum of leptons and jets, missing transverse mass, pseudorapidity for leptons and jets, invariant mass for lepton pairs and jet pairs, and the transverse momentum for leading and subleading jets.

```
27.0 = ptl
35.0 = ptj
20.0 = misset
2.5 = etal
4.5 = etaj
20.0 = mml1
500.0 = mmjj
65.0 = ptj1min
35.0 = ptj2min
```

Event Pass Rates Cut	LL		LT		TT		Unpol	
	Count	Rate	Count	Rate	Count	Rate	Count	Rate
Total	100000	1.00	100000	1.00	100000	1.00	100000	1.00
Lepton cut	26793	0.27	27563	0.28	30021	0.30	28388	0.28
MET cut	23816	0.24	25036	0.25	27755	0.28	25979	0.26
Jet cut	19002	0.19	20602	0.21	22352	0.22	21116	0.21

Cross Sections	LL (pb)		LT (pb)		TT (pb)		Unpol (pb)	
Total	0.000505		0.002843		0.004288		0.007653	
Lepton cut	0.000135		0.000784		0.001287		0.002173	
MET cut	0.000120		0.000712		0.001190		0.001988	
Jet cut	0.000096		0.000586		0.000959		0.001616	

Table 2: Event pass rates and cross sections after pre-cut in MadGraph for LL, LT, TT, and Unpolarized samples.

In table 2, we can see that applying pre-cutting in MadGraph improves the event throughput, which increases the efficiency of data utilization. Furthermore, the change in cross-section after each cut is exactly the ratio times the original cross-section.

2.2 Transformation from pp -cmf to WW -cmf

Since the kinematic distribution in the paper[1] is under WW -cmf, we should consider to transform our original generated events in MadGraph from pp -cmf to WW -cmf.

2.2.1 Work Flow

To transform the cmf of the events, we have to first extract the 4-vector from the original events including leptons, neutrinos, and W bosons, where $W = l + \nu_l$. Then, we calculate the WW system kinematic by:

$$P_{WW} = P_{W+} + P_{W-}, \quad (9)$$

where P is the 4-vector.

To move into the WW -cmf, we must apply a Lorentz transformation to boost our target particles in the opposite direction of the WW system's motion. Therefore, we boost using $-\vec{\beta}$.

$$\vec{\beta}_{WW} = \frac{\vec{p}_{WW}}{E_{WW}}, \quad (10)$$

where $\vec{p}_{WW}$ equals (p_x, p_y, p_z) .

Since we only know $(E_\ell, p_{\ell,x}, p_{\ell,y}, p_{\ell,z})$ and $(p_{\nu,x}, p_{\nu,y})$, we have to calculate the $p_{\nu,z}$ and E_ν . By using the W -mass constraint, which assumes the W boson is perfectly on-shell,

$$(p_\ell + p_\nu)^2 = m_W^2, \quad (11)$$

where $m_W \approx 80.4$ GeV.

By expanding equation 11 using the known lepton kinematics and the measured transverse missing energy, we can end up with a standard quadratic equation for the neutrino's longitudinal momentum and then figure out the $p_{\nu,z}$ and E_ν . However, the transverse part of the momentum is non zero, therefore, the boost is not a purely longitudinal boost, but a 3D boost.

2.2.2 Expectations after the Transformation

- $\Delta\eta_{\ell\ell}$:

$$\Delta\eta' \neq \Delta\eta, \quad (12)$$

The difference in pseudorapidity should change on an event-by-event basis. Because a 3D boost mixes the momentum components, there is no single constant shift applied to the leptons. The $\Delta\eta$ distribution in the boosted frame should look smeared or broaden compared to the lab frame.

- m_T :

$$m_{T,pp-cmf} \rightarrow M_{WW,WW-cmf}, \quad (13)$$

Since we calculated the longitudinal momentum for neutrino, we can then figure out the invariant mass, instead of the transverse mass. While the total momentum in WW -cmf is zero, the invariant mass of the system is simply the total energy of all decay products:

$$M_{WW} = E_{\ell 1}^* + E_{\nu 1}^* + E_{\ell 2}^* + E_{\nu 1}^*. \quad (14)$$

- $\Delta\phi_{jj}$:

When analyzing the Vector Boson Scattering (VBS), the leading and subleading tagging jets have a very specific kinematic signature: they are highly forward. Which means they have massive total energies and longitudinal momenta. The $\gamma E\beta_T$ term becomes massive and so, the $\Delta\phi_{jj}$ distribution is will be heavily distorted.

$$\vec{p}'_{Ti} = \vec{p}_{Ti} + \left[\frac{\gamma - 1}{\beta^2} (\vec{\beta} \cdot \vec{p}_i) - \gamma E_i \right] \vec{\beta}_T. \quad (15)$$

2.2.3 Results

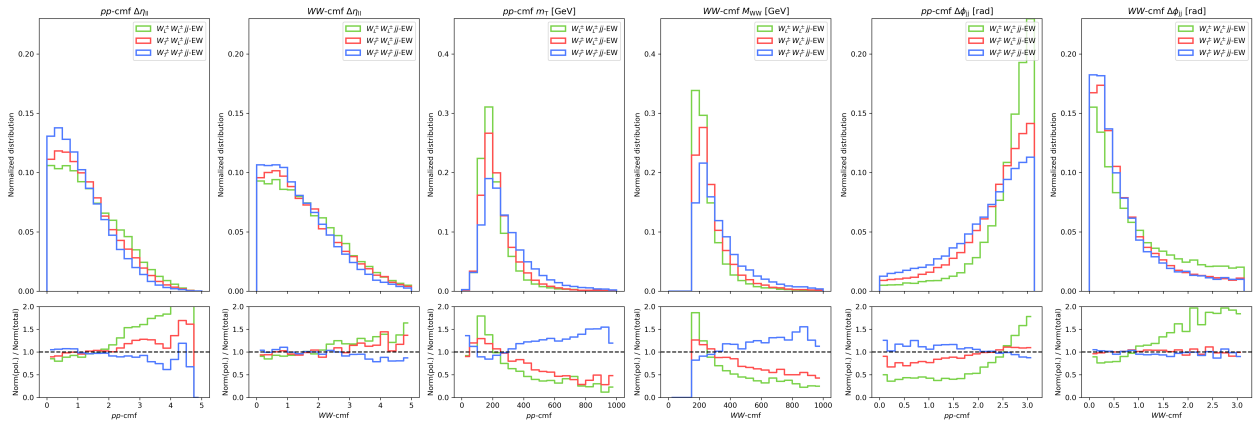


Figure 2: Distribution in the signal region after pre-cut and transformation to the WW -cmf in MadGraph for the difference in pseudorapidity between leading and subleading leptons, the transverse mass, and the difference in azimuthal angle between leading and subleading jets.

- **Peak values for $\Delta\eta_{\ell\ell}$ decreases:** The decrease in the TT peak for $\Delta\eta_{\ell\ell}$ in the WW -cmf occurs because boosting into the center-of-mass frame removes the global longitudinal “push” of the lab frame, unfolding the system and allowing the naturally broader angular distributions of transversely polarized bosons—which follow a $(1 \pm \cos^2 \theta^*)$ distribution—to spread out and lower the normalized peak. Furthermore, because TT leptons are emitted more longitudinally, they are highly sensitive to “smearing” and potential under-boosting caused by the algebraic approximation of p_z , which further flattens the distribution compared to the LL state.
- **M_{WW} goes to zero below ≈ 160 GeV:** The M_{WW} values drop to zero below approximately 160 GeV primarily due to the on-shell threshold; since we force a W -mass constraint of 80.385 GeV for each reconstructed boson, the total invariant mass cannot physically fall below the sum of those two masses (~ 161 GeV). Additionally, the selection criteria—specifically the high p_T requirements for leptons and the $m_{jj} > 500$ GeV VBS topology cut—effectively filter out low-energy phase space where a lower M_{WW} might otherwise occur.
- **$\Delta\phi_{jj}$ goes to zero:** In pp -cmf, the $\Delta\phi_{jj}$ distribution is governed by the hard scatter and recoil dynamics. But in WW -cmf, the distribution is heavily destroyed by the massive Energy term through the 3D boost. The jets are forced to point in similar transverse direction, making them highly collinear in ϕ .

3 March 8th

3.1 WW -cmf in MadGraph

```
generate p p > w+{0} w+{0} j j QCD=0, w+ > l+ vl
add process p p > w-{0} w-{0} j j QCD=0, w- > l- vl~
output sample/script/LL
launch sample/script/LL
```

```
shower = Pythia8
detector = Delphes
analysis= OFF
madspin= OFF
```

```
set run_card nevents 100000
set run_card ebeam1 6500
set run_card ebeam2 6500
```

```
set run_card me_frame 3,4,5,6
```

```
set run_card cut_decays True
set run_card ptj 34.0
set run_card ptl 26.0
set run_card miset 29.0
```

```
set run_card etaj 4.6
set run_card etal 2.6
```

```
set run_card mmjj 450
```

```
set run_card ptj1min 64
set run_card ptj2min 34
```

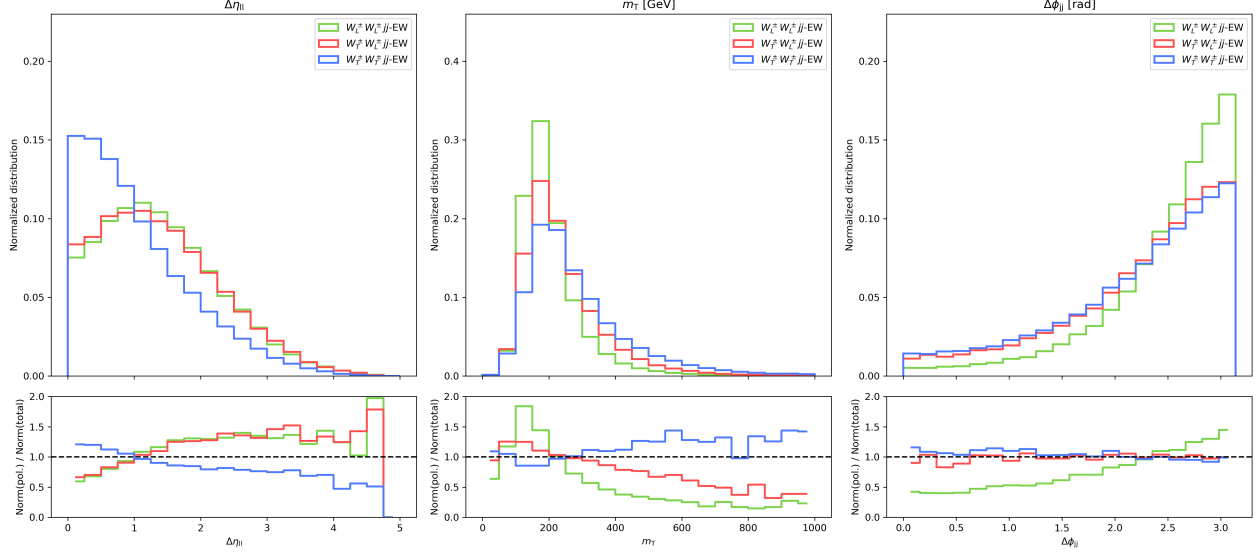


Figure 3: Distribution in the signal region after pre-cut in the WW -cmf in MadGraph for the difference in pseudorapidity, transverse mass, and difference in azimuthal angle.

pt _i -pt _l tbl/cell/end args int=0Γ	pp -cmf (fb)				WW -cmf (fb)			
	LL	LT	TT	Total	LL	LT	TT	Total
Total	0.505	2.843	4.288	7.653	0.396	1.263	2.334	3.913
Lepton cut	0.135	0.784	1.287	2.173	0.201	0.650	1.284	2.082
MET cut	0.120	0.712	1.190	1.988	0.186	0.611	1.218	1.966
Jet / VBS cut	0.096	0.586	0.959	1.616	0.148	0.484	0.960	1.547
Jet cut (Thesis)	0.177	1.030	1.718	2.925	0.277	0.874	1.744	2.896
Fraction	5.9%	35.7%	58.4%	100%	9.3%	30.4%	60.3%	100%
Fraction (Thesis)	6.1%	35.2%	58.7%	100%	9.6%	30.2%	60.2%	100%

Table 3: Comparison of cutflows, cross-sections, and polarization fractions for pp -cmf and WW -cmf frames. Results from local MadGraph generations are compared against the thesis references. All cross-sections are expressed in fb.

Table 3 compares the cutflow and polarization results of the local MadGraph generation against the reference thesis. While the absolute cross-sections in the signal region (Jet/VBS cut) are roughly 45% lower than the thesis values, the polarization fractions show excellent agreement. Specifically, the LL, LT, and TT fractions deviate by less than 1% from the reference in both the pp -cmf and WW -cmf frames. This confirms that despite the normalization difference, the underlying angular distributions and physics of the polarized states are correctly modeled. Figure 3 also shows the underlying kinematic distributions are similar to the results in the thesis.

3.2 Sherpa3

In Sherpa3, instead of generating an event that is 100 % purely "LL" or 100 % "TT", Sherpa generates a physical event (a specific arrangement of outgoing jets and leptons) and then calculates a weight for each polarization state for that exact same event.

Also, we must separate the W^+W^+jj and W^-W^-jj processes when using the Pol_Cross_Section feature because of how the generator strictly handles memory allocation for event weights.

3.2.1 Sample Preparation

For W^+ :

```
EVENTS: 150000          #number of events to generate

BEAMS: [2212, 2212]     #incoming colliding particles; 2212 for a proton
BEAM_ENERGIES: 6500     #incoming particle energy for each

#specifies the internal structure model of the proton to be used during the collision
PDF_LIBRARY: LHAPDFSherpa    #tell Shepra to interface with the library
PDF_SET: PDF4LHC21_40_pdfas  #highly accurate PDF set

#93: any light quark, anti-quark, or gluon
#24: w+ boson
PROCESSES:
#2 incoming partons scatter to produce 2 same sign bosons and 2 outgoing partons
- 93 93 -> 24 24 93 93:
#forbidding any strong-force interactions in the hard matrix element
    Order: {QCD: 0, EW: 4}

SELECTORS:
#invariant mass of the 2 outgoing partons must be between 100-13,000 GeV
- [Mass, 93, 93, 100.0, 13000.0]

PARTICLE_DATA:
#treat w+ boson as strictly on-shell during the initial matrix element calculation
24:
    Width: 0.0
#turns on the decay module for heavy particles
HARD_DECAYS:
    Enabled: true
#crucial!!
#ensures the spin-correlations and polarization of w boson are calculated and passed
down to their decay products
    Pol_Cross_Section:
        Enabled: true
        Reference_System: COM      #COM:ww-cmf; Lab: pp-cmf
#disables specific decay routes for the w boson
#status : 0 meaning turns a decay channel off.
    Channels:
```

```

    '24,2,-1': {Status: 0}          #up+anti-down quark (hardronic decay)
    '24,4,-3': {Status: 0}          #charm+anti-strange quark (hardronic decay)
    '24,16,-15': {Status: 0}        #tau neutrino+anti-tau lepton (hardronic decay)
#dictates how and where the final simulated events are saved
EVENT_OUTPUT: HepMC3_GenEvent[Sherpa_Wp]

```

For W^- :

```
EVENTS: 150000
```

```
BEAMS: [2212, 2212]
```

```
BEAM_ENERGIES: 6500
```

```
PDF_LIBRARY: LHAPDFSherpa
```

```
PDF_SET: PDF4LHC21_40_pdfas
```

```
PROCESSES:
```

```

- 93 93 -> -24 -24 93 93:
  Order: {QCD: 0, EW: 4}

```

```
SELECTORS:
```

```
- [Mass, 93, 93, 100.0, 13000.0]
```

```
PARTICLE_DATA:
```

```

-24:
  Width: 0.0

```

```
HARD_DECAYS:
```

```
Enabled: true
```

```
Pol_Cross_Section:
```

```
Enabled: true
```

```
Reference_System: COM
```

```
Channels:
```

```

'-24,-2,1': {Status: 0}
'-24,-4,3': {Status: 0}
'-24,-16,15': {Status: 0}

```

```
EVENT_OUTPUT: HepMC3_GenEvent[Sherpa_Wm]
```

	Cross Sections (fb)				Event Pass Rates	
	LL	LT	TT	Unpol	Raw Events	Fraction
Total	1.239	3.781	6.874	11.915	299,998	1.0000
Lepton cut	0.577	1.710	3.230	5.479	142,376	0.4746
MET cut	0.477	1.473	2.767	4.658	120,052	0.4002
Jet cut	0.239	0.753	1.398	2.349	49,259	0.1642

Table 4: Cross sections for different polarization states and corresponding event pass rates at each selection cut stage.

According to table 4, the event pass rate for Sherpa simulation is much smaller than MadGraph due to less cuts in the event generating step. Perhaps we can explore whether there are other cuts can be added to Sherpa to enhance the event usage.

	Sherpa pp -cmf (fb)				Sherpa WW -cmf (fb)			
	LL	LT	TT	Total	LL	LT	TT	Total
Total	1.060	4.787	6.083	11.915	1.254	3.816	6.873	11.915
n_{ll} cut	0.472	2.180	2.906	5.456	0.575	1.714	3.214	5.456
MET cut	0.381	1.845	2.516	4.635	0.469	1.470	2.755	4.635
VBS cut	0.175	0.915	1.306	2.335	0.237	0.749	1.392	2.335
VBS cut (Thesis)	0.182	1.072	1.732	2.933	0.285	0.909	1.778	2.933
Fraction	7.3%	38.2%	54.5%	100%	10.0%	31.5%	58.5%	100%
Fraction (Thesis)	6.1%	35.9%	58.0%	100%	9.6%	30.6%	59.8%	100%

Table 5: Comparison of cutflows, cross-sections, and polarization fractions for pp -cmf and WW -cmf frames using Sherpa. Results from local generations are compared against the thesis Sherpa ($W^\pm W^\pm jj$) references.

Table 5 shows that the calculated cross-sections are in close agreement with the thesis references. Furthermore, the polarization fractions for the LL, LT, and TT states remain consistent with the reference values across both the pp -cmf and WW -cmf frames.

4 March 12

4.1 Sherpa3 Rerun

It was identified that the initial analysis in section 3.2 lacked a proper detector simulation stage; specifically, Delphes was not applied to the Sherpa-generated events. Furthermore, the standalone Python implementation for jet reconstruction was unreliable for this analysis. Consequently, the Sherpa output files were re-processed through Delphes to ensure a more robust reconstruction of jets within the signal region.

Table 6 summarizes the results following this detector simulation. Notably, the discrepancy persists, which aligns with the results from MadGraph: the cross-sections obtained after the three signal region cuts are approximately 50% lower than those reported in the reference thesis. Despite this difference in absolute magnitude, the relative polarization fractions (LL, LT, and TT) remain broadly consistent with the thesis values.

	Sherpa3 + Delphes pp -cmf (fb)				Sherpa3 + Delphes WW -cmf (fb)			
	LL	LT	TT	Total	LL	LT	TT	Total
Total	1.060	4.787	6.083	11.915	1.254	3.816	6.873	11.915
n_{ll} cut	0.265	1.245	1.740	3.202	0.332	0.985	1.905	3.202
MET cut	0.217	1.057	1.515	2.744	0.275	0.847	1.645	2.744
VBS cut	0.112	0.585	0.870	1.540	0.154	0.479	0.922	1.540
VBS cut (Thesis)	0.182	1.072	1.732	2.933	0.285	0.909	1.778	2.933
Fraction	7.1%	37.4%	55.5%	100%	9.9%	30.8%	59.3%	100%
Fraction (Thesis)	6.1%	35.9%	58.0%	100%	9.6%	30.6%	59.8%	100%

Table 6: Comparison of cutflows, cross-sections, and polarization fractions for pp -cmf and WW -cmf frames using Sherpa after Delphes detector simulation. Local generations are compared against the thesis Sherpa ($W^\pm W^\pm jj$) references.

4.2 PDF Selection

I was using the preset settings for event generation in MadGraph of PDF selection:

```
nn23lo1 = pdlabel
230000 = lhaid
```

But to compare the results with Sherpa, we need to use the same PDF. Therefore, I decided to use the following settings:

In Sherpa3:

```
PDF_SET: PDF4LHC21_40_pdfas
```

In MadGraph:

```
set pdlabel lhapdf
set lhaid 93300
```

And I found that the cross section for different PDF changed:

	LL (fb)		LT (fb)		TT (fb)	
	PDF	Nominal	PDF	Nominal	PDF	Nominal
Total	0.434	0.396	1.372	1.263	2.531	2.334
n_{ll} cut	0.221	0.201	0.702	0.649	1.376	1.284
MET cut	0.205	0.186	0.660	0.611	1.302	1.218
VBS cut	0.163	0.148	0.518	0.484	1.017	0.960

Table 7: Comparison of cross-sections for different polarization states showing the effect of the PDF selection. The “PDF” columns represent the new PDF choice, while “Nominal” represents the baseline generation. All values are in fb.

To ensure consistency between MadGraph and Sherpa3, the default NNPDF2.3 LO (lhaid 230000) was replaced with the modern PDF4LHC21 benchmark (lhaid 93300).

As shown in table 7, updating to the PDF4LHC21 set results in a systematic increase in cross sections across all polarization states. This shift is expected, as the modern PDF set incorporated more precise experimental data and updated theoretical constraints compared to the decade-old nominal set. This transition ensures that the local simulations align with current LHC standards and provide more reliable results.

4.3 Particle-level phase space

After I read the thesis in detail again, I found out that the thesis was comparing the cross section at particle-level, not the detector-level. However, the previous cross section for my simulation were calculated at the detector-level. Instead of rerunning the event generations, we can use the free Delphes GenJet branch. Despite Delphes is a detector simulator, its default configuration actually runs the FastJet clustering algorithm on the PYTHIA truth particles before it applies any detector smearing.

- Particle-Level jets: Must run a jet clustering algorithm on all stable final state particles (Status = 1). (anti- kt algorithm with $R = 0.4$).
- Particle-Level MET: This is calculated as the negative vector sum of the transverse momenta of all stable, weakly interacting particles.
- Particle-Level Leptons: These are stable electrons and muons straight from the PYTHIA output.

```
particles = "Particle.PT", "Particle.Eta", "Particle.Phi", "Particle.Mass"...
genjets = "GenJet.PT", "GenJet.Eta", "GenJet.Phi", "GenJet.Mass"
genmet = "GenMissingET.MET"
```

Sample	WW -cmf			pp -cmf		
	LL	LT+TL	TT	LL	LT+TL	TT
Thesis MadGraph	0.277 fb 9.58%	0.874 fb 30.18%	1.744 fb 60.24%	0.177 fb 6.07%	1.030 fb 35.20%	1.718 fb 58.73%
Local MadGraph	0.256 fb 9.59%	0.814 fb 30.56%	1.595 fb 59.85%	0.166 fb 6.19%	0.954 fb 35.60%	1.560 fb 58.22%
Thesis Sherpa $W^\pm W^\pm jj$	0.285 fb 9.60%	0.909 fb 30.59%	1.778 fb 59.81%	0.182 fb 6.09%	1.072 fb 35.91%	1.732 fb 58.00%
Local Sherpa3	0.252 fb 9.87%	0.780 fb 30.58%	1.519 fb 59.54%	0.192 fb 7.47%	0.977 fb 37.93%	1.406 fb 54.60%

Table 8: Comparison of cross-sections and polarization fractions for WW -cmf and pp -cmf frames. The reference thesis values (MadGraph and Sherpa) are compared against the local MadGraph generations and local Sherpa3 generations at the particle-level VBS cut.

Table 8 provides a comprehensive comparison between local simulation and the reference thesis for both pp -cmf and WW -cmf. The polarization fractions show similar results. While the absolute cross section in the local generations are slightly lower than the thesis reference, which may likely due to higher order corrections or PDF sets. This result also shows that the underlying physics modeling and symmetry factors in MadGrpah or Sherpa3 setups are correctly aligned with the thesis results.

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5.1 Background events

Following are the ranked background events mentioned in [2]:

1. $W^\pm Z$ background (dominant)
2. Data-driven backgrounds (~ 18 % combined):
 - Non-prompt leptons: arises from leptons originating from hadron decays and jets that are misidentified as leptons. The source of these events is mainly W +jets and semi-leptonic $t\bar{t}$ processes.
 - Charge-flip background (electron charge misidentification): This background primarily comes from $Z \rightarrow e^+e^-$ decays. A high-energy electron emits a hard photon via bremsstrahlung, which then undergoes pair production into a high-energy positron and a soft electron. The soft electrons get lost in the detector, and the misidentified opposite-charged positron is measured instead, incorrectly passing the same-charge requirement.
3. QCD $W^\pm W^\pm jj$ background (~ 5.2 %)
4. Small prompt backgrounds (< 2 % combined)
5. Interference $W^\pm W^\pm jj$ –INT (~ 1.6 %)

5.1.1 $W^\pm Z jj$ background

- $pp \rightarrow W^\pm Z jj \rightarrow \ell^\pm \nu \ell^\pm \ell^\mp$ (including both pure electroweak $W^\pm Z$ –EW and QCD-induced $W^\pm Z$ –QCD scattering)
- Event generator:
 - $W^\pm Z$ –EW: Madgraph
 - $W^\pm Z$ –QCD: Sherpa2.2.12 + data-driven approach (will Shepra3 simulation be sufficient?) + control region restrictions
- CMF: evaluated in the WW -cmf and pp -cmf
- PDF set: NNPDF3.0_nnlo_as.0118

This background originates from the fully leptonic decay. It mimics the signal region criteria when one of the leptons from the Z boson decay fails baseline selection or falls completely outside the detector’s acceptance. This leaves exactly two same-charged leptons, two jets, and missing transverse momentum. It consists of both pure electroweak and QCD-induced scattering events.

Comprises the electroweak scattering $W^\pm Z$ –EW and the strong scattering $W^\pm Z$ –QCD. $W^\pm Z$ –EW can be simulated by MadGraph well, while $W^\pm Z$ –QCD is known to be poorly modeled by Monte Carlo simulations. Sample generated from Sherpa for $W^\pm Z$ –QCD still needs to be corrected using the same data-driven approach as in [2].

5.1.2 QCD $W^\pm W^\pm jj$ background

- $pp \rightarrow \ell^\pm \nu \ell^\pm \nu jj$ via strong interaction vertices at order $O(\alpha_{\text{EW}}^4 \alpha_S^2)$
- Event generator: MadGraph
- CMF: evaluated in the WW -cmf and pp -cmf
- PDF set: NNPDF3.0_nnlo_as_0118

This is an irreducible background that yields the exact same final state particles as the signal, but it is produced at the order $O(\alpha_{\text{EW}}^4 \alpha_S^2)$ involving strong QCD interaction vertices.

5.1.3 Interference background $W^\pm W^\pm jj$ –INT

- Quantum mechanical interference between the purely electroweak and QCD-induced $W^\pm W^\pm jj$ diagrams that share identical external quarks, calculated at order $O(\alpha_{\text{EW}}^5 \alpha_S)$
- Event generator: MadGraph and Sherpa

For MadGraph, the interference is calculated by:

$$N_{\text{int}} = N_{\text{unpol}} - \sum_{\text{pol}} N_{\text{pol}} \quad (16)$$

For Sherpa, the interference contribution is directly provided by an event weight as for the individual polarization states.

- CMF: evaluated in the WW -cmf and pp -cmf
- PDF set: modeled by MadGraph using NNPDF3.0

5.2 Control Region

5.2.1 $W^\pm Z$ control region

- Two signal leptons and an additional third signal lepton with $p_T > 15$ GeV
- In the case of two electrons they have to fulfill $|\eta| < 1.37$ and $|m_{ee} - m_Z| > 15$ GeV
- At least two leptons satisfy $m_{\ell\ell} \geq 20$ GeV
- $m_{\ell\ell\ell} \geq 106$ GeV
- $p_T^{\text{miss}} \geq 30$ GeV
- At least two signal jets with the highest $p_T > 65$ GeV and the second highest $p_T > 35$ GeV
- $m_{jj} \geq 200$ GeV
- $|\Delta y_{jj}| > 2$
- No jet is b -tagged using the DL1r tagger with the 85% efficiency working point

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6.1 Signal events

Adjusting the PDF set for MadGraph into

```
set pdlabel lhpdf
set lhaid 260000
```

which sets directly the same PDF set used in the thesis (NNPDF30_nlo_as_0118)

The following table is the after the correct PDF set selection with the thesis (NNPDF30_nlo_as_0118). We can compare how different overlap removal affect the cross section. But it isn't clearly mentioned in the thesis how the overlap removal is applied.

MadGraph	with step 5	without step 5	corrected	thesis
LL	0.250	0.218	0.229	0.277
LT	0.794	0.662	0.699	0.874
TT	1.558	1.193	1.309	1.744

Table 9: Signal events generated from MadGraph with the PDF set: NNPDF30_nlo_as_0118 and with different overlap removal conditions.

where the column “corrected” is set by the rules mentioned in the thesis 6.3.4 Fiducial Signal Region. If the resulting jet overlaps with an electron within $\Delta R < 0.4$, the electron is removed for $p_{T,e}/p_{T,jet} < 0.5$. The jet is removed if the electron carries more than 50% of the transverse jet momentum.

And the step 5 is a action to remove jets with $R > 0.3$ of the signal leptons.

Sherpa3	with step 5	without step 5	corrected	thesis
LL	0.252	0.233	0.236	0.285
LT	0.780	0.700	0.717	0.909
TT	1.519	1.245	1.325	1.778

Table 10: Signal events generated from Sherpa3 with the PDF set: PDF4LHC21_40_pdfas and with different overlap removal conditions.

We can see from both table that different choice of the overlap removal affect the cross section very much. However, in both the thesis or in the references mentioned in the thesis didn't clearly denote what overlap removal is applied to the signal events. Therefore, the discrepancy between our simulation result and the result in the thesis might due to the choice of the overlap removal.

6.2 Background events

I found some tables and figures that we might be able to compare with the background events before training the DNN model. There are different kinds of cut mentioned in the thesis, which is summarized in the following figure: According to this figure 4, we can compare our simulation result with table G.2-5 in the thesis [3] which includes the number of signal and background events in SR, CR, and low m_{jj} region. Also, it includes both the pp -cmf and WW -cmf. Therefore, by the following equation,

$$N_{\text{expected}} = \sigma \times \mathcal{L} \times \epsilon \quad (17)$$

Requirement	SR	Low- m_{jj} CR	WZ CR
Leading and subleading lepton p_T		> 27 GeV	
Electron $ \eta $	< 2.47 (1.37 in ee), excluding $1.37 \leq \eta \leq 1.52$		
Muon $ \eta $		< 2.5	
Leading (subleading) jet p_T		> 65 (35) GeV	
Additional jet p_T		> 25 GeV	
Jet $ \eta $		< 4.5	
$m_{\ell\ell}$		> 20 GeV	
E_T^{miss}		> 30 GeV	
Charge misid. $Z \rightarrow ee$ veto	$ m_{ee} - m_Z > 15$ GeV		–
b -jet veto	$N_{b\text{-jet}} = 0, p_T^{b\text{-jet}} > 20$ GeV, $ \eta^{b\text{-jet}} < 2.5$		
$N_{\text{veto leptons}}$	$= 0$	$= 0$	$= 1, p_T > 15$ GeV
$m_{\ell\ell\ell}$	–	–	> 106 GeV
m_{jj}	> 500 GeV	$200 < m_{jj} < 500$ GeV	> 200 GeV
$ \Delta y_{jj} $		> 2	

Figure 4: Summary of the event selection for the signal region and control region.

- N_{expected} : number of expected events
- σ : the theoretical cross-section of the process
- $\mathcal{L}$: the integrated luminosity of the data (for the full ATLAS Run 2 dataset, this is 139 fb^{-1})
- ϵ : the efficiency and acceptance. (the event pass rate)

we might be able to validate our results before applying the DNN training.

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7.1 Background simulation

7.1.1 Simulation code

We generate the following background events by MadGraph 3.7.0 [4] and all of the PDF sets for each background simulation are selected to be `lhaid 260000`, which is `NNPDF30_nlo_as_0118`. For $W^\pm W^\pm jj$ QCD

```
generate p p > l+ vl l+ vl j j QCD=2 QED=4
add process p p > l- vl~ l- vl~ j j QCD=2 QED=4
```

For $W^\pm Z$ QCD

```
generate p p > l+ l- l+ vl j j QCD=0
add process p p > l+ l- l- vl~ j j QCD=0
```

For $W^\pm Z$ EW

```
generate p p > l+ l- l+ vl j j QCD=2 QED=4
add process p p > l+ l- l- vl~ j j QCD=2 QED=4
```

7.1.2 Results

Table 11: Comparison of signal region cross sections (in fb) between simulation results, Stange Thesis [1] (Table G.2), and ATLAS paper [2] (Table 3). Expected yields from the thesis and paper have been converted to cross sections using the integrated luminosity mentioned in both paper of 139 fb^{-1} .

Process	MadGraph3.7.1 [fb]	Sherpa3 [fb]	Thesis [1][fb]	Paper [2][fb]
Signal Processes ($W^\pm W^\pm jj$ EW)				
$W_L W_L$ (LL)	0.132		0.145	–
$W_L W_T$ (LT)	0.426		0.453	–
$W_T W_T$ (TT)	0.847		0.922	–
sum	1.405	1.353	1.520	–
WW unpolarized	1.378		1.554	1.691
Background Processes				
WZ -EW	0.150	0.409	0.108	0.107
WW -QCD	0.093	0.138	0.175	0.173
WZ -QCD	0.375	0.252	0.554	0.597

The signal region cuts obey the summarized table 2 in paper [2], but adding the detail that lepton pairs should be in same sign. We can see from table 11 that for electroweak processes (both signal and background), MadGraph [4] simulated a great result which is consistence with the reference. But for QCD-induced results, MadGraph [4] does not generate results reasonably. Therefore, we will try to generate some background events from Sherpa[5] to examine the consistency of the result. We can also see that for events generated by Sherpa[5], QCD-induced WW background behaves better than MadGraph, while MadGraph generates better for all electroweak events. However, there is still some inconsistency for the WZ -QCD induced background.

Table 12: Expected yields in the signal region (SR). The ATLAS results are shown alongside Thesis values and our simulation estimates using different generator versions, assuming an integrated luminosity of 139 fb^{-1} .

Process	MG 3.7.0	Sherpa 3.0.3	ATLAS	Thesis 9.1	Thesis G.2
$W^\pm W^\pm jj$ EW	195.31	188.03	235	206.52	211.29
$W^\pm W^\pm jj$ QCD	12.95	19.12	24	24.05	24.27
$W^\pm Z jj$ EW	20.81	56.98	14.9	14.95	15.07
$W^\pm Z jj$ QCD	52.13	34.99	83	82.75	77.0

We can also compare the result by the expected yields in the signal region shown in table 12. Which shows the same trend that Sherpa generates better WW -QCD induced events than MadGraph, while MadGraph generates better electroweak processes ($WWjj$ and $WZjj$) than Sherpa.

However, both event generators exhibit low consistency for the $WZjj$ -QCD induced process.

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I took the following steps to try to generate results consistent with Feng-Yang's:

- **Cross section calculation for Sherpa:** Instead of extracting the nominal cross section from the `hepmc` file, we summed the cross sections for the positively charged processes (e.g., W^+) and the negatively charged processes (e.g., W^-), and then multiplied the total by the branching ratio.
- **Delphes `run_card`:** I initially used the ATLAS card but found that it lacked the fat jet finding configuration inside the `run_card` script. Therefore, I switched to the default `run_card` originally provided with Delphes and altered the jet radius to 0.4 to align with the paper.
- **Narrow width approximation (NWA) instead of the full process:** Because there were issues generating the full process in Sherpa (e.g., excessive integration time), we switched to using the narrow width approximation (NWA), which was also used for generating the signal process in the paper. Additionally, we adjusted the events generated by MadGraph to use NWA instead of the full process.

Table 13: Expected yields in the signal region (SR). The ATLAS results are shown alongside Thesis values and our simulation estimates using different generator versions, assuming an integrated luminosity of 139 fb^{-1} .

Process	MG 3.7.0	Sherpa 3.0.3	ATLAS	Thesis 9.1	Thesis G.2
$W^\pm W^\pm jj$ EW	201.89	169.26	235	206.52	211.29
$W^\pm W^\pm jj$ QCD	15.20	17.59	24	24.05	24.27
$W^\pm Zjj$ EW	21.94	32.05	14.9	14.95	15.07
$W^\pm Zjj$ QCD	50.84	78.35	83	82.75	77.0

Comparing this table with last week's, we observe that for MadGraph, the WW processes are now closer to the reference values. For Sherpa, the expected yield for the WW processes has decreased, which might be related to the different Delphes `run_card` used. For the WZ processes, Sherpa demonstrates better consistency than last week's results.

To make further comparisons with the results from the thesis or paper, we need to determine the exact Delphes `run_card` that was used. As our latest results indicate, the choice of `run_card` significantly affects the cross section, as shown in the following table:

Process	Expected Events (139 fb ⁻¹)	
	delphes_card_default.tcl	delphes_card_ATLAS.tcl
WWjj-EW-NWA	201.89	212.46
WWjj-QCD-NWA	15.20	15.86
WZjj-EW-NWA	21.94	24.82
WZjj-QCD-NWA	50.84	53.26

Table 14: Comparison of expected event yields across four processes, evaluated using the ATLAS Delphes card versus the Default Delphes card.

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9.1 Summary of DNN Training Input Variables

Table 15 summarizes the variables used as inputs for the neural network training. To improve the training process for the DNN algorithms, some variables are pre-scaled to achieve a more Gaussian-distributed input.

The η values of leptons and jets can be used to define the lepton Zeppenfeld variable:

$$Z_l^* = \left| \frac{\eta_l - \frac{1}{2}(\eta_{j1} + \eta_{j2})}{\eta_{j1} - \eta_{j2}} \right|. \quad (18)$$

This is a measure of how central a lepton is relative to the two tagging jets.

The definition of the massless early-projected transverse mass is:

$$m_{o1}^{ll,MET} = \sqrt{(p_T^{l1} + p_T^{l2} + p_T^{miss})^2 - |\vec{p}_T^{l1} + \vec{p}_T^{l2} + \vec{p}_T^{miss}|^2}, \quad (19)$$

This is an advanced kinematic approximation used to estimate system masses involving missing transverse momentum, assuming massless visible products.

And the mass-preserving transverse mass is defined by:

$$m_T^{ll,MET} = \sqrt{(E_T^{ll} + p_T^{miss})^2 - |\vec{p}_T^{l1} + \vec{p}_T^{l2} + \vec{p}_T^{miss}|^2}, \quad (20)$$

Or generally, it can be defined for object x , for x being the leading lepton, sub-leading lepton, or the de-lepton system:

$$m_T(x, E_T^{miss}) = \sqrt{(E_T^x + E_T^{miss})^2 - |\vec{p}_T^x + \vec{p}_T^{miss}|^2}, \quad (21)$$

where $E_T = \sqrt{m^2 + |p_T|^2}$.

9.2 Dataset separation

The training data is split into train validation, and test sets according to the event number in a ration of 3:1:1.

- The test set: $((\text{EventNumber} - i_{\text{fold}}) \bmod 5) = 0$

Kinematics	Description	Scaling
Low-level variables		
$p_T^{\ell_1}, p_T^{\ell_2}, p_T^{j_1}, p_T^{j_2}$	p_T of the leading/subleading lepton/jet	$\log_{10}(x)$
$\eta^{\ell_1}, \eta^{\ell_2}, \eta^{j_1}, \eta^{j_2}$	η of the leading/subleading lepton/jet	x
ℓ_1 type, ℓ_2 type	Flavor of the leading/subleading lepton	x
$\phi^{\ell_2} - \phi^{\ell_1}, \phi^{j_1} - \phi^{\ell_1}, \phi^{j_2} - \phi^{\ell_1}$	Redefined ϕ of the subleading lepton/ leading jet/ subleading jet	x
p_T^{miss}	Missing transverse momentum	$\log_{10}(x)$
$\phi(p_T^{\text{miss}}) - \phi^{\ell_1}$	Redefined ϕ of the missing transverse energy	x
High-level variables		
$Z_{\ell_1}^*, Z_{\ell_2}^*$	Zeppenfeld variable of the leading/subleading lepton	$\sqrt{x}$
$m_T^{\ell_1, \text{MET}}, m_T^{\ell_2, \text{MET}}$	Transverse mass of the respective lepton and p_T^{miss}	$\sqrt{x}$
$\Delta R_{\ell\ell}, \Delta R_{jj}$	ΔR separation between the two leptons / two jets	x
$\Delta\eta_{\ell\ell}$	Pseudorapidity separation between the two leptons	$\sqrt{x}$
$m_{\ell\ell}, m_{jj}$	Invariant mass of the dilepton / dijet system	$\log_{10}(x)$
$p_T^{\ell\ell}$	Transverse momentum of the dilepton system	$\sqrt{x}$
$m_T^{\ell\ell, \text{MET}}$	Transverse mass of the dilepton system	$\sqrt{x}$
$m_{o1}^{\ell\ell, \text{MET}}$	Early-projected massless invariant mass	$\sqrt{x}$
Δy_{jj}	Rapidity separation between the two jets	x
$\Delta\phi_{jj}$	Azimuthal separation between the two jets	x
$(p_T^{\ell_1} \cdot p_T^{\ell_2}) / (p_T^{j_1} \cdot p_T^{j_2})$	p_T ratio of the lepton and jet systems	$\log_{10}(x + 0.02)$
$\min(\Delta R_{\ell_i, j_k})$	Minimal ΔR separation between any lepton and jet	x

Table 15: Summary of DNN input variables and their pre-scaling functions.

- The validation set: $((\text{EventNumber} - i_{\text{fold}} + 1) \bmod 5) = 0$
- The training set: the remaining events

The index i_{fold} defines the different splits used for cross-validation in the hyperparameter optimization.

9.3 Training process

The training processes are divided into two parts: First, an inclusive network designed to separate the electroweak (EW) signal from all backgrounds. Second, the polarization-specific networks trained exclusively on the EW samples to distinguish specific polarization states. In details, it is trained to separate longitudinal single-boson polarization (LL and LT v.s. TT) or double-boson polarization (LL v.s. LT and TT) from the remaining transverse polarization states.

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10.1 5-Fold Method

The k -fold method is a data-splitting and cross-validation strategy used to reliably evaluate neural network performance and prevent biases when making predictions.

10.1.1 Data Categorization

The dataset is divided into three distinct and independent categories, using the i_{Fold} index:

- Test dataset: 20% with $((\text{EventNumber} - i_{\text{Fold}}) \bmod 5) = 0$. Kept completely hidden from the training and optimization process to test the final model.
- Validation dataset: 20% with $((\text{EventNumber} - i_{\text{Fold}} + 1) \bmod 5) = 0$. Used to evaluate the network's performance and tune hyperparameter during training.
- Training dataset: 60%, which is the remaining events that are not test or validation datasets. Used to adapt and optimize the network's trainable parameters.

where the index $i_{\text{Fold}} \in [0, 1, 2, 3, 4]$ and independent of the physical properties of the events.

10.1.2 Cross-Validation and Application

- Training Multiple Networks: The process iteratively trains five separate networks. Each network uses a different overlapping combination of the training folds and evaluates against non-overlapping validation and test sets.
- Evaluating Performance: Training across different independent validation datasets allows the author to calculate a mean network performance and estimate its statistical uncertainty.
- Preventing Mismodeling Bias: Neural networks naturally perform better on the data they were trained on than on unseen data. If predictions evaluated on training data were compared against measured data, it would introduce a systematic bias.
- Final Unbiased Predictions: To guarantee unbiased predictions across the entire dataset, each event's final prediction is generated exclusively by the specific DNN that held that event in its unseen test dataset.

10.1.3 Explanation of Proper Nouns

- **Unseen data** in the context of machine learning and the k -fold method refers specifically to the test dataset, which is used to provide an objective test of how well the final model can generalize to new, real-world examples. It is data that is kept completely hidden from the neural network while it is learning and optimizing its parameters.

$$s_j = \sum_i x_i w_{ij} + \theta_j \quad (22)$$

- **Weights** determine the strength and importance of the connection between two neurons. Mathematically, a weight is a multiplier. The output of a previous neuron (x_i) is multiplied by its assigned weight (w_{ij}) before it enters the next neuron.
- **Bias** (θ_j) is an additional parameter added to the neuron, independent of the input data. Mathematically, bias acts as an offset. By adding a bias, the network shifts the activation function left or right, determining how high the sum of the weighted inputs needs to be before the neuron activates.

10.2 Network Architecture

Based on section 9.2 in the reference thesis [3], the neural networks used are exclusively fully-connected feed-forward neural networks. The architecture is structured as follows:

- **Normalization Layer (Input):** Applies a scaling function to the input variables, normalizing their distributions to have a mean of 0 and a variance of 1. This ensures all inputs have similar scales without losing their kinematic phase space information.
- **Repeating Hidden Sequence:**

1. **Dense Layer:** A standard fully-connected layer.
2. **Batch Normalization Layer:** Used to normalize the output from the preceding dense layer.
3. **Activation Layer:** Applies the activation function, specifically the Swish function defined as:

$$f_{\text{Swish}}(x) = \frac{x}{1 + e^{-x}} \quad (23)$$

For a binary classification the Sigmoid function:

$$f_{\text{Sigmoid}}(x) = \frac{1}{1 + e^{-\beta x}} \quad (24)$$

4. **Dropout Layer:** Introduced for regularization to prevent overtraining.
- **Output Layer:** The repeating sequence concludes with a final dense layer, which computes the ultimate output score of the network.

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11.1 Ap plying Simulated Data to Real Measured Data

Based on the thesis, there are three primary methods and strategies used to bridge the gap between Monte Carlo (MC) simulated training data and real measured data.

1. Preventing Mismodeling Bias via the k -Fold Method

Neural network naturally perform better on the data they were trained on than on unseen data. If a network’s prediction on its familiar training data was directly compared to real measured data, it would introduce an artificial systematic mismodeling (bias). The detailed explanation of this strategy is in Section 5.8 and Section 12.2.

To guarantee unbiased predictions across the entire dataset, each event’s final classification score is generated exclusively by the specific dense-NN that that held that event in its completely unseen “test” dataset.

2. Validation Strategy using Control Regions

Before deploying the neural network on the actual final measurement (the signal region), it is necessary to check if the simulation matches the real data. However, testing the dense-NN directly in the signal region could influence the final measurement and introduce a human “analysis strategy bias”. The definition of the control region is in Section 6.3.2 and the results of the validation tests are in Section 10.2.

Validation tests are safely performed in designated “control regions”. These are phase spaces that are kinematically similar to the signal region but are background-only. By comparing the dense-NN output on real measured data vs. simulated data in these control regions, we can ensure the simulation matches reality before looking at the actual signal.

*Run trained DNN on the simulated background data and real measured data in control region and plot a histogram of the output scores (0(Background) to 1(Signal)). If it behaves correctly, two histograms should stack perfectly on top of each other and should assign mostly low scores to both datasets.

3. Correcting Simulation Errors with Data-Driven Backgrounds

Simulations are never perfect replicas of real life and fail to accurately predict certain detector artifacts, such as fake leptons or electrons flipping their charges. This approach is detailed in Section 7.2.2 and Section 7.2.3.

Since MC simulation models these backgrounds very poorly, the analysis uses data-driven techniques. This involves extracting metrics like “fake factor” directly from real-world measured data to overwrite or correct the flawed simulation predictions.

In summary, they first train five independent model on perfectly labeled simulated data. Then, prove the simulation matches reality in a safe control region and fix the broken parts (like fake-leptons)of the simulation using data-driven factors. Finally, the measured data is strictly evaluated by the specific model that corresponds to that fold. (If the real `EventNumber % 5` equals 2, that event is evaluated only by Model 2.)

11.2 References to Compare with

The thesis [3] contains specific figures that are highly valuable for validating our script, which is listed as follow:

1. Input Variable Distributions

In Appendix C, there are several figures of the pre-processed physical variables. We can plot these variables to verify the data parsing, variable calculations, and data shape match the thesis exactly.

2. Dense-NN Output Score Distributions

Once our model is trained, we can run it on the test dataset. We can plot our output scores using an equidistant binning of exactly 20 bins, as this is the standard binning convention used in the thesis for optimizing and visualizing Neural Network outputs. We can compare the distribution of the final classification probability scores outputted by the Sigmoid layer of our trained Dense-NN with Section 10.2.

3. ROC-AUC Performance Curves

We can compare our final testing AUC scores against the reported AUC scores in the thesis to confirm our network architecture and hyperparameters are performing at the expected efficiency, which can be found in Chapter 9.

11.3 ML Workflow

11.3.1 Data Pre-processing

Feng-Yang and me decided to use `parquet` as the dense-NN input variable file format. And the workflow should be as follows:

1. Extract Raw Data:
2. Calculate Variables: For each event, calculate all the low-level and high-level variables listed in Table 9.2 in the thesis [3].
3. Apply Pre-scaling: Apply the scaling mentioned in the thesis [3] in Table 9.2 to let the data resemble a Gaussian distribution for models learning more effectively.
4. Assign Folds: Calculate the physical Event Number and use the formula:

$$((\text{Event Number} - i_{\text{Fold}}) \bmod 5) = 0 \quad (25)$$

to tag each event with a fold number (0, 1, 2, 3, or 4).

5. Write to Parquet

12 5/17

12.1 Foundational ML Concepts

1. The **loss Curve** is a line graph that tracks how many “mistakes” the neural network is making overtime. In our script, the network uses Binary Cross-Entropy (BCE) loss, which is standard for binary classification. The lower the loss, the better the network is at distinguishing target from background.

2. An **epoch** is one complete cycle where the neural network has seen the entire training dataset exactly once. Because datasets in particle physics are massive, the data is fed to the network in smaller chunks called “batches.” (If we have 10,000 events and a batch size of 1,000, it takes 10 batches to get through all the data. Once all 10 batches have passes through the network, one epoch is complete.)

Reading it a second time (Epoch 2) helps understand and retain the material better, steadily lowering the loss.

3. **AUC** stands for area under the curve, and it specifically refers to the receiver operating characteristic (ROC) curve. The ROC-AUC evaluates the separation power of the model.

12.2 dense-NN Result

For the initial test, I am using 3592 events for the $\text{DNN}_{W^\pm W^\pm}$ task and 6679 events for the DNN_{pol} task. The whole set of input features are used for the training. And the dense-NN structure for the two model is specified as follow:

```
history_ww = train_5_fold(
    df=df_ww,
    features_cols=features_cols,
    num_layers=8,
    neurons_per_layer=491,
    dropout_rate=0.355,
    lr=0.002,
    model_name="DNN_WW"
)

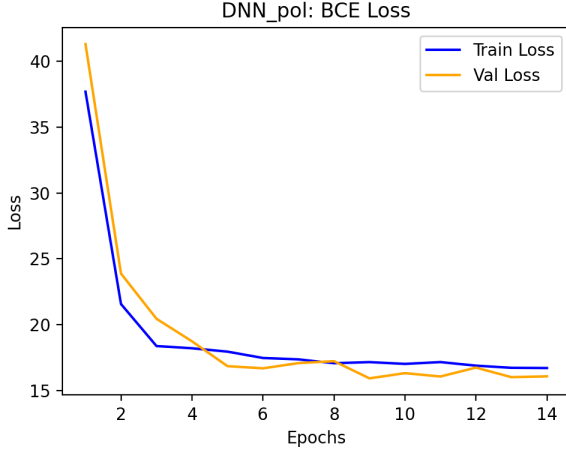
history_pol = train_5_fold(
    df=df_pol,
    features_cols=features_cols,
    num_layers=9,
    neurons_per_layer=296,
    dropout_rate=0.381,
    lr=0.0015,
    model_name="DNN_pol"
)
```

Classification Task	Validation AUC	Test AUC
$\text{DNN}_{W^\pm W^\pm}$	0.7654 ± 0.0107	0.7601 ± 0.0163
DNN_{pol} (LL vs. TX)	0.7878 ± 0.0128	0.7865 ± 0.00083

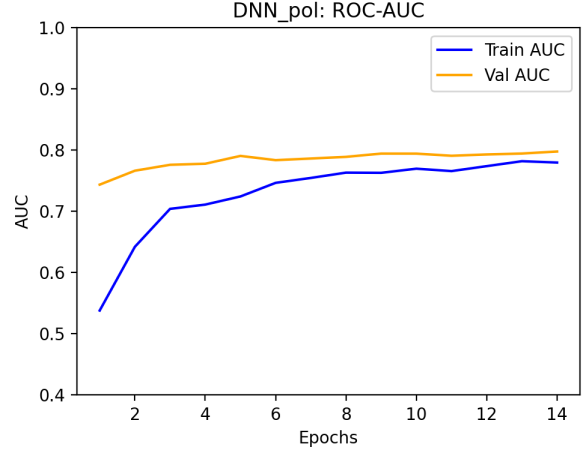
Table 16: Classification performance for DNN models using shared data.

Figure 5 illustrates the evolution of the loss and AUC curve. The quantitative performance of the models across the 5-fold ensemble is summarized in table 16.

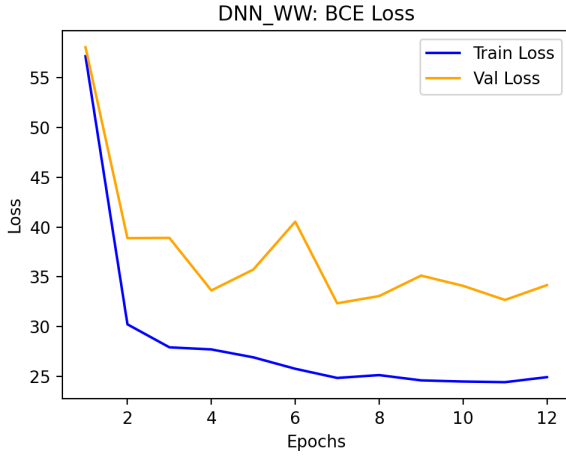
As anticipated for a preliminary test, the training curves indicate that the models—particularly DNN_{WW} —have not yet fully converged, as evidenced by the volatility in validation loss. The next steps will involve generating larger training samples to bridge the generalization gap, increasing the



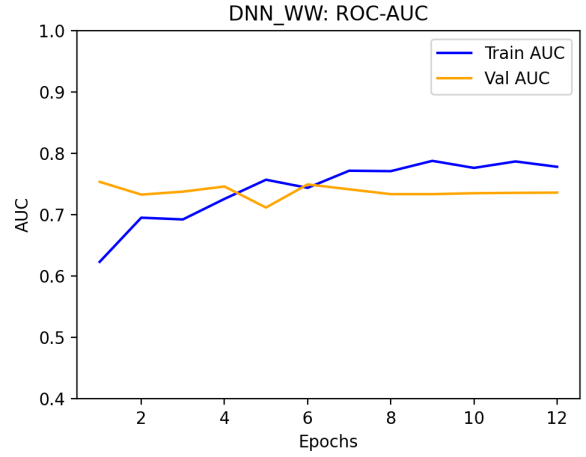
(a) DNN_{pol}: BCE Loss



(b) DNN_{pol}: ROC-AUC



(c) DNN_{WW}: BCE Loss



(d) DNN_{WW}: ROC-AUC

Figure 5: Model performance comparison: BCE Loss (left column) and ROC-AUC (right column) for DNN_{pol} and DNN_{WW}.

number of epochs to observe long-term stability, and testing optimized hyperparameters to improve the ROC-AUC plateau.

*Table 9.1 in thesis [3] outlines the Class Weights. By multiplying the signal by a massive number so that the signal event looks just as loud and important as the background events.

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13.1 Performance of the Dense-NN Model

The performance for models using larger number of dataset are shown in figure 6 and table 17.

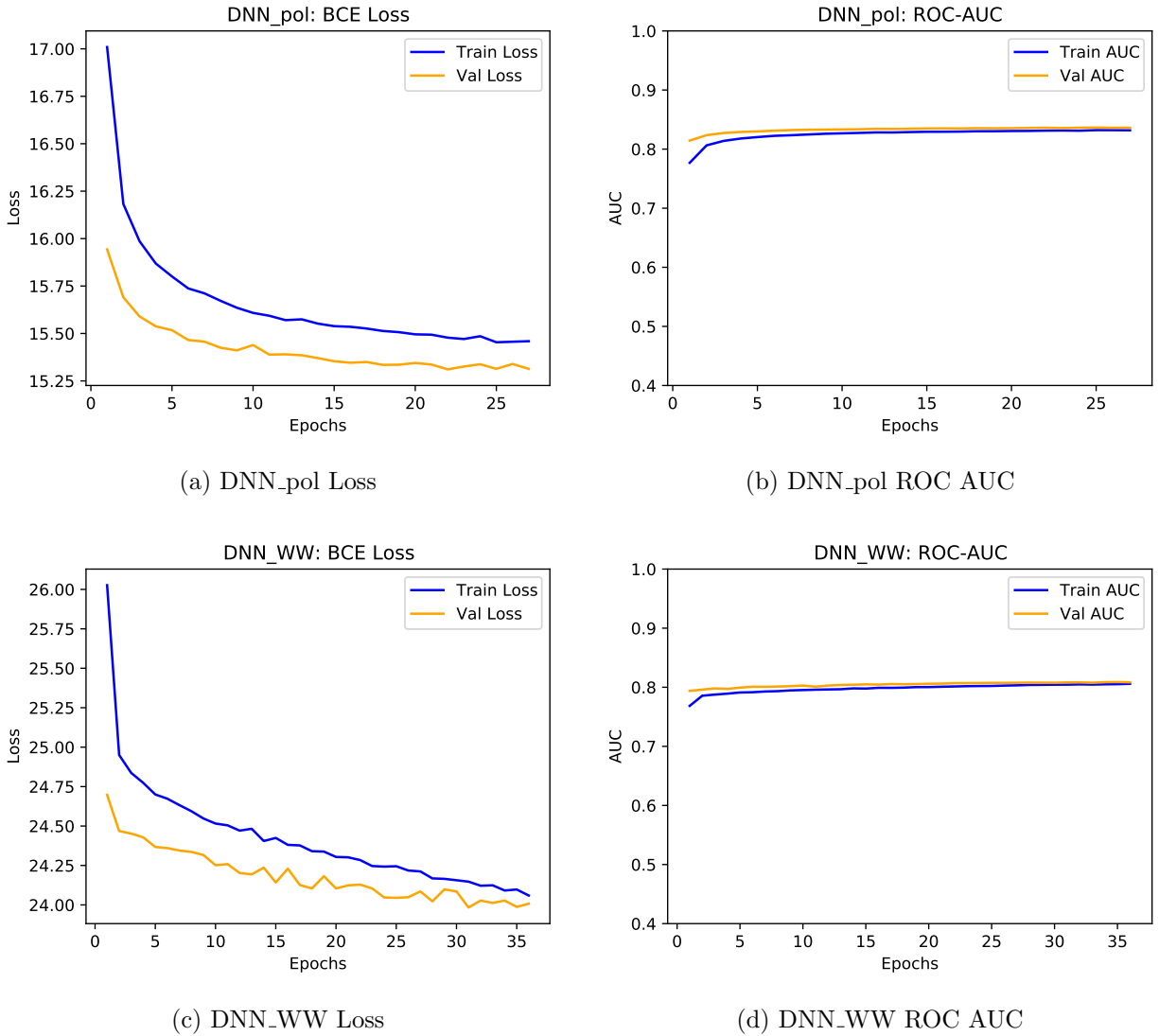


Figure 6: Performance metrics for DNN_pol and DNN_WW models.

- DNN_pol Model:

- Loss: Both Training and Validation loss have decreased compared to last week's results, showing a more consistent downward trend. Last week, the training loss appeared

higher and less converged, while the current curves show a clearer stabilization as epochs progress.

- ROC AUC: The AUC values are showing healthier behavior. The model now converges to an AUC around 0.8 with less volatile gaps between the training and validation sets, suggesting better generalization than the previous week.
- DNN _{$W^\pm W^\pm$} Model:
 - Loss: There is a notable improvement in the stability of the training loss curve. While last week’s validation loss showed significant fluctuations, the current results indicate that the loss is better behaved and the model is finding a more stable minimum.
 - ROC AUC: The training and validation AUC curves have become more tightly coupled, which is a positive indicator that the training process is becoming more reliable and less prone to overfitting.

Classification Task	Validation AUC	Test AUC
DNN _{$W^\pm W^\pm$}	0.8067 ± 0.0013	0.8065 ± 0.0013
DNN _{pol} (LL vs. TX)	0.8383 ± 0.0013	0.8381 ± 0.0011

Table 17: Classification performance for DNN models using 1 million dataset

Table 17 also shows higher AUC values for both the validation and testing for DNN _{$W^\pm W^\pm$} and DNN_{pol} compared to last week’s result.

13.2 Code Explanation

- `model.train()` is used to switch the model to train mode, which will cause layers such as Dropout (randomly dropping neurons) and BatchNorm (batch normalization) to operate as they did during training. (e.g., Dropout will initiate random dropping, and BatchNorm will use the statistic of the current batch and update the internal mean/variance). Calling `model.train()` during training ensures these layers work correctly; during validation or testing, `model.eval()` should be used instead.
- `optimizer.zero_grad()`: Clears the accumulated gradients on the model parameters. Each time `loss.backward()` is executed, PyTorch adds the computed gradients to the parameter’s `.grad` file; if `zero_grad()` is not called first, the new gradients will be superimposed on the old gradients, leading to incorrect parameter updates. A typical training flow is:
 1. `optimizer.zero_grad`: clears gradients.
 2. `outputs = model(inputs)`: forward propagation.
 3. `loss = criterion(outputs, targets)`: calculates loss.
 4. `loss.backward()`: backward propagation, specifying `.grad`.
 5. `optimizer.step()`: updates parameters.
- `criterion = nn.BCELoss(reduction='none')`: returning a per-sample loss tensor. Binary cross-entropy for one sample (prediction p , label $y \in \{0, 1\}$) is:

$$\text{BCE}(p, y) = -y \log p - (1 - y) \log(1 - p) \quad (26)$$

`reduction='none'` means the loss call returns an elementwise tensor of shape (batch,1) so the code can multiply each sample's loss by its event weight and then aggregate manually.

- Usage of `Weight`:
 1. Multiplying the per-sample loss by `Weight` makes the training objective care more about heavy-weight events and less about light-weight events. In other words, the model is trained to minimize the weighted expected loss, not the plain unweighted average.
 2. `roc_auc_score(y_true, y_pred, sample_weight=weights)`: Weighted ROC-AUC means each sample contributes to the ROC curve proportionally to its weight.

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14.1 Initial State and Final State Polarization

In section 3.3.3 in the thesis [3], they mentioned that at high center-of-mass energy, the final state $W_L^\pm W_L^\pm$ originates almost exclusively from the $W_L^\pm W_L^\pm$ initial state. Other $VV \rightarrow V_L V_L$ processes are significantly affected by helicity flips, but for same-charged $W^\pm W^\pm \rightarrow W^\pm W^\pm$ scattering at the high center-of-mass energy achieved at the LHC, the polarization of the final state is similar to the initial state polarization. Therefore, the measurement of $W^\pm W^\pm \rightarrow W_L^\pm W_L^\pm$ at the LHC is almost equivalent to the measurement of $W_L^\pm W_L^\pm \rightarrow W_L^\pm W_L^\pm$.

14.2 Paper [6]

14.2.1 The Dataset and Goal

- Process: parton-level simulations of Vector Boson Fusion: $uu \rightarrow W^+ W^- uu$.
- Decay: fully hadronic decays: $W^+ \rightarrow u\bar{d}$ and $W^- \rightarrow \bar{u}d$.
- The Classification Goal: theory-informed tagger to identify the LL polarization final states against the background combinations: LT, TL, and TT.

14.2.2 The Methodology

The author explicitly designs the system so that the machine learning model does not act as a black-box classifier.

1. Agnostic Training:

The reinforcement learning algorithm is trained on an unpolarized sample of the $W^+ W^- uu$ events. During this phase, the reward function is a mixture of all four matrix elements. The agent learns the optical particle assignments completely blind to the actual WW polarizations.

2. Interpretable Mapping:

The sole job of the neural network is to predict the correct mapping between the observed final-state particles and the partons entering the matrix element.

3. Theory-Informed Scoring:

Once the network outputs the optimal particle mapping, the classification score is not generated by the neural network. Instead, it is computed directly using the exact analytic matrix elements for the different polarizations.

The optimal classification score is defined as a likelihood ratio:

$$L(\tilde{x}_i) = \frac{m_{LL}(\tilde{x}_i)}{m_{LT}(\tilde{x}_i) + m_{TL}(\tilde{x}_i) + m_{TT}(\tilde{x}_i)} \quad (27)$$

Because this score is computed directly from exact tree-level amplitudes, it is fully interpretable and respects all physical symmetries of the process.

14.2.3 The Results

The paper demonstrates that this theory-informed tagger performs extremely well without relying on traditional black-box classification:

- **Mass Reconstruction:**

Using the mappings predicted by the neural network, the reconstructed invariant masses of the W bosons are in very good agreement with the true mass distribution (shown in Figure 7 of the paper).

- **Tagging Performance (ROC/AUC):**

The classifier’s performance for isolating the LL signal was evaluated using ROC curves (Figure 8):

- Background consists of TT only events: AUC = 0.925
- Background consists of full mix of LT, TL, and TT events: AUC = 0.858

The author concludes that this approach works highly effectively in practice because RL algorithm learns the correct topological assignments without needing prior knowledge of the polarization states.

15 6/5

15.1 Chapter 12 in Thesis

15.1.1 Machine Learning Output and Binning

In the search for the longitudinal $W^\pm W^\pm$ scattering signal, the analysis first relies on Deep Neural Networks (DNNs) to separate the signal from the background. The output is sorted into bins (e.g., Bin 1 to Bin 10), where higher bins are “highly likely to be Signal.” In Chapter 12 of the thesis, this corresponds to the two-dimensional bin optimization algorithm designed to maximize the expected significance by isolating regions with the highest $W^\pm W^\pm jj$ -EW purity.

15.1.2 Theoretical Expectations and Null Hypothesis

Before unblinding the data, theoretical expectations for the signal and background must be established.

- **Expectations:** The number of expected events is calculated using the sum of weights, which accounts for the base cross-section, lepton efficiencies, trigger efficiencies, and kinematic corrections.
- **Null Hypothesis:** The null hypothesis assumes that no longitudinal signal exists, setting the signal strength parameter to zero ($\mu_L = 0$). Under this hypothesis, the expected yield consists purely of the background expectation, subject to statistical fluctuations ($\pm\sqrt{\text{Background}}$).

15.1.3 Unblinding and the Fit Algorithm

Upon “unblinding,” the real measured data in the signal region is revealed. To extract the signal significance, a “Fit Algorithm” is applied. Chapter 12 defines this as a *profile likelihood fit*. The fit adjusts the expected distributions to the measured data by varying the normalizations ($\mu_L, \mu_T, \mu_{WZ\text{-}QCD}$) and constraining systematic uncertainties via nuisance parameters (θ, γ).

15.1.4 Significance and Realities

A simplified significance formula:

$$Z = \frac{\text{Real Data} - \text{Background Expectation}}{\sqrt{\text{Background Expectation}}} \quad (28)$$

In the thesis, this is calculated using an asymptotic formula based on the profile likelihood ratio test statistic q_0 , where the discovery significance is $Z_0 = \sqrt{q_0}$.

For example, there might two possible outcomes (Realities):

- **Reality A (Low Significance):** $Z = 0.5\sigma$, indicating just a background fluctuation with no signal found. This closely matches the actual observed result for the fully longitudinal (LL) polarization state in the WW -cmf, which yielded an observation significance of only 0.082σ due to a statistical fluctuation in the data. Meaning no signal is found.
- **Reality B (High Significance):** $Z = 5\sigma$, indicating a successful signal discovery. While the thesis did not reach the 5σ discovery threshold, the measurement for the combined longitudinal and transverse (LX) state achieved an observed significance of 3.4σ , providing strong evidence for the process.

15.2 DNN Score Distribution

15.2.1 The Corrected Weighting Method

- **Absolute Weights:** The thesis [3] states: “To avoid negative loss values in the BCE loss calculation, the absolute values of the Monte Carlo event weights $|\omega_{MC,i}|$ are used for the training.”
- **Class Balancing:** Table 9.1 in the thesis states: “In the training, the signal is scaled to the background yield.” The logic `scaling_factor = sum_bkg / sum_signal` and multiplying it solely to the signal events.

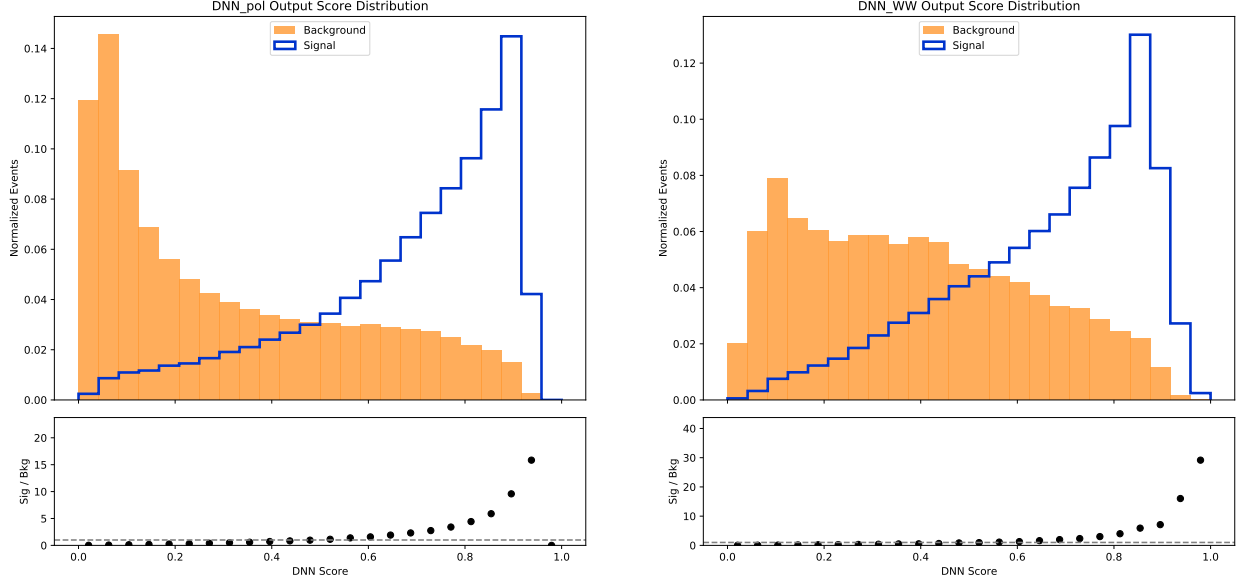
15.2.2 Result

Both neural networks are showing strong predictive capability. According to figure 7a and figure 7b, they both demonstrate a clear, healthy separation between the Signal and Background distributions. The networks have successfully learned the kinematic features necessary to push background events toward a score of 0.0 and signal events toward 1.0.

16 6/20

16.1 Discovery Significance and 95% CL Upper Limit

To establish a more comprehensive comparison with the reference thesis, the expected discovery significances (Z_0) and the expected 95% confidence level (CL) upper limits were also calculated.



(a) Polarization score distribution

(b) WW score distribution

Figure 7: DNN score distributions for polarization and WW processes.

16.1.1 Theory and Methodology

1. The Binned Likelihood Function

Before calculating any significance, the analysis constructs a likelihood function, $\mathcal{L}$. This function describes the probability of observing the data given a specific signal strength (μ) and a set of nuisance parameters (θ):

$$\mathcal{L}(\mu, \theta) = \prod_{j=1}^{N_{\text{bins}}} \frac{(\mu s_j(\theta) + b_j(\theta))^{n_j}}{n_j!} e^{-(\mu s_j(\theta) + b_j(\theta))} \prod_{k=1}^{N_{\text{sys}}} C(\theta_k) \quad (29)$$

- μ : The signal strength modifier ($\mu = 1$ is the Standard Model expectation, $\mu = 0$ is the background-only hypothesis).
- $s_j(\theta)$ and $b_j(\theta)$: The expected signal and background yields in bin j , which vary based on the nuisance parameters.
- n_j : The actual observed data events in bin j .
- $C(\theta_k)$: The constraint functions (usually Gaussian or log-normal distributions) that restrict the nuisance parameters within their known systematic uncertainties.

2. The Profile Likelihood Ratio

To test different hypotheses, the likelihood function is converted into a **Profile Likelihood Ratio** (Λ). This isolates the signal strength μ by “profiling out” (maximizing over) all the nuisance parameters:

$$\Lambda(\mu) = \frac{\mathcal{L}(\mu, \hat{\hat{\theta}})}{\mathcal{L}(\hat{\mu}, \hat{\theta})} \quad (30)$$

- **Denominator** ($\mathcal{L}(\hat{\mu}, \hat{\theta})$): The unconditional maximum likelihood. It is found by letting both μ and θ float to whatever values best fit the data.
- **Numerator** ($\mathcal{L}(\mu, \hat{\theta})$): The conditional maximum likelihood. It is found by locking μ to a specific test value (like 0 for a discovery test) and letting only θ adjust to best fit the data.

3. Calculating Discovery Significance (Z_0)

To claim a discovery, the analysis tests the background-only hypothesis ($\mu = 0$).

The Test Statistic (q_0):

$$q_0 = \begin{cases} -2 \ln \Lambda(0) & \text{for } \hat{\mu} \geq 0 \\ 0 & \text{for } \hat{\mu} < 0 \end{cases} \quad (31)$$

(The condition $\hat{\mu} < 0$ is set to zero because a downward fluctuation of background should not be counted as evidence for a new particle).

The p -value and Significance: Using asymptotic formulas (Wilks' theorem), the probability of the background fluctuating to mimic the signal (the p -value) is calculated from q_0 . The discovery significance (Z_0) is then extracted using the inverse of the standard Gaussian cumulative distribution function (Φ^{-1}):

$$Z_0 = \Phi^{-1}(1 - p_0) \approx \sqrt{q_0} \quad (32)$$

4. Calculating the 95% CL Upper Limit (CL_s Method)

Because the observed significance for the longitudinal scattering was too low, the analysis transitions to setting an upper limit on μ . ATLAS uses the **CL_s method** to prevent setting artificially tight limits in regions where the background simply fluctuated downwards.

The Test Statistic for Limits (q_μ): To test a specific signal strength μ , a different test statistic is used:

$$q_\mu = \begin{cases} -2 \ln \Lambda(\mu) & \text{for } \hat{\mu} \leq \mu \\ 0 & \text{for } \hat{\mu} > \mu \end{cases} \quad (33)$$

The CL_s Calculation: The algorithm calculates two p -values based on q_μ :

- p_μ : The probability of the signal plus background hypothesis being compatible with the data.
- $1 - p_b$: The probability of the background-only hypothesis being compatible with the data.

The CL_s value is their ratio:

$$\text{CL}_s = \frac{p_\mu}{1 - p_b} \quad (34)$$

5. The Final Limit:

To find the 95% Confidence Level upper limit (μ_{95}), computational solvers (like RooFit/Minuit) iteratively scan and test different values of μ until they find the exact signal strength where:

$$\text{CL}_s = 0.05 \quad (35)$$

This process requires intensive numerical minimization across thousands of dimensions (one for every bin and systematic uncertainty), which is why it is far more complex than a standard algebraic formula.

16.1.2 Results

We can compare these findings with Table 12.4 and Table 12.9 in the thesis. The compiled results are presented in Table 18.

	Expected Significance (Z_0)		Expected 95% CL Upper Limit	
	Thesis	My Model	Thesis	My Model
LL (μ_{LL})	1.512 σ	1.4929 σ	2.273	1.3769
LX (μ_{LX})	4.557 σ	4.3881 σ	—	—

Table 18: Expected discovery significances (Z_0) and expected 95% CL upper limits for the LL and LX polarization states in the WW center-of-mass frame.

As shown in Table 18, the expected discovery significances derived from the current model closely track the thesis baseline for both the longitudinal-longitudinal (LL) and longitudinal-transverse (LX) polarization states, yielding 1.49 σ and 4.39 σ respectively.

Our model establishes an expected 95% CL upper limit for the LL signal strength of 1.38, compared to the thesis limit of 2.27. The primary reason our limit is significantly tighter is that our calculation is currently a 'Stat-Only' fit (including only a flat 10% background normalization uncertainty). The thesis limit of 2.27 was derived using a 'Full Systematics' fit that profiled hundreds of real-world detector uncertainties (such as Jet Energy Scale, Muon Tracking resolution, and fake-lepton background variations) across the binned kinematic distributions. Because our neural network is currently unhindered by these complex detector systematics, the profile likelihood fitter operates with much higher confidence, resulting in a mathematically more stringent limit. This proves our baseline DNN kinematics are separating the shapes correctly, and the limit will naturally inflate to match the thesis once full detector systematics are applied.

16.2 SPANet (Symmetry-Preserving Attention Networks)

SPANet [7, 8] is a deep learning architecture designed specifically to solve the jet assignment problem.

16.2.1 The Jet Assignment Problem

In collider physics, when heavy particles are created, they rapidly decay into variable-size sets of observed particles or jets. When processing large-scale physical simulations, extracting signals from background noise requires correctly matching these detected final-state jets back to their parent particles.

- Event reconstruction has traditionally been performed by evaluating every possible permutation of jet assignments. However, the combinatorial background scales exponentially with the number of final-state jets—a particularly acute challenge when analyzing complex multi-jet signatures.
- Furthermore, these arbitrary ordering methods ignore the natural physical symmetries inherent in the decays.

16.2.2 How it Works

1. Data Representation: The Permutationless Input

Traditional algorithms require jets to be ordered arbitrarily before being fed into the system, creating artificial sequence dependence. SPANet eliminates this by treating an event as an unordered point cloud.

- **The input:** An event is represented as a set of N jets. Each jet is a feature vector containing kinematic and qualitative data (like b -tagging scores), forming a matrix $X \in \mathbb{R}^{N \times D}$.
- **Event-level encoding:** SPANet passes this matrix through a Transformer encoder using multi-head self-attention. This translates the raw data into a contextualized mathematical representation.

2. Symmetric Tensor Attention (STA)

Once the jets are encoded, the network must assign them to parent particles. Instead of outputting a single jet at a time, SPANet generates a joint probability distribution over all possible combinations using STA. STA outputs a rank- k tensor $P \in \mathbb{R}^{N \times N \times \dots \times N}$, where $P(j_1, \dots, j_k)$ represents the network’s predicted probability that a specific combination of jets originates from the parent particle.

Crucially, STA enforces jet symmetry by ensuring the learnable weight tensors are commutative across the symmetric axes, such that $P(j_1, j_2) = P(j_2, j_1)$.

3. Global Event Topology: Combined Symmetric Loss (CSL) Function

While STA handles symmetries within a single particle’s decay, CSL manages symmetries between identical parent particles.

During training, the network calculates the prediction loss for all valid permutations of identical parent particles. It then applies the minimum achievable loss and updates the weights. This prevents the network from penalizing itself for guessing the correct jet assignment but arbitrarily swapping the labels of two indistinguishable parent particles.

4. Final Output

For every expected particle defined in the event topology, SPANet produces two distinct outputs:

- **Detection Probability (DP):** A scalar value indicating the network’s confidence that the parent particle is fully reconstructible within the event.
- **Assignment Distribution (AD):** The rank- k tensor generated by the STA layer.

Once the DP crosses a certain threshold, the network extracts the coordinates of the peak value from the AD tensor. Those coordinates represent the final, physically-symmetric jet assignment for that particle.

17 6/28

17.1 Fully Hadronic Process

We found a thesis: Precision Measurements of Vector Boson Scattering Production and Searches for Charged Higgs Bosons at the Large Hadron Collider [9] that is related to the “fully hadronic” process. The signal topologies and categorization considered in the analysis:

- One leg targets the $V \rightarrow qq$ ($V = W, Z$) decay in both the boosted and resolved decay products.
- The other leg targets three different decay modes:
 - Invisible: $Z \rightarrow \nu\nu$
 - Boosted $Z \rightarrow b\bar{b}/c\bar{c}$
 - Boosted $V \rightarrow q'q'$

where q denotes a quark of any possible type, while q' for light quarks (used) only.

Considering both of the two, total of 6 categories are supposed to be identified. However, due to the overwhelming of QCD multijet background, the two all-visible resolved ($V_r Z_b \rightarrow qq + b\bar{b}/c\bar{c}$ and $V_r V_b \rightarrow qq + q'q'$) categories are not considered. Only four categories are considered which are defined as follow:

- Boosted-Boosted (B-B): events with two boosted V bosons both decaying to two light quarks”

$$VV \rightarrow q'q' + q'q'$$

- Boosted-Invisible (B-MET): event with one invisibly decaying Z boson and one boosted Z boson decaying into two quarks:

$$ZV \rightarrow \nu\nu + qq$$

- Resolved-Invisible (R-MET): events with one invisibly decaying Z boson and one resolved V boson decaying to two quarks.

$$ZV \rightarrow \nu\nu qq$$

- Boosted-WithB (B-Btag): events with two boosted V bosons where at least one is a Z boson decaying to heavy-flavor quarks.

$$VZ \rightarrow qq + b\bar{b}/c\bar{c}$$

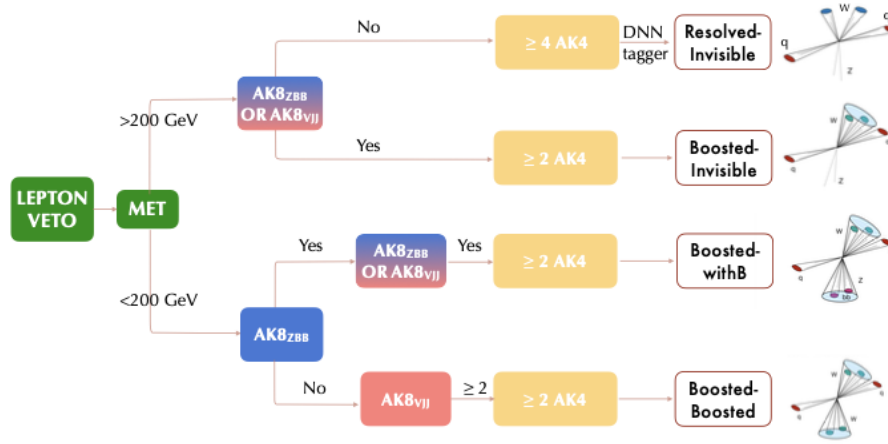


Figure 8: The event selection flow of the four categories mentioned in the thesis.

- $AK8_{ZBB}$: This is an AK8 fat jet specifically identified as a Z boson decaying into heavy-flavor quarks ($b\bar{b}$ or $c\bar{c}$). It requires specific heavy-flavor tagging criteria.
- $AK8_{Vjj}$: This is an AK8 fat jet identified as a Vector Boson (W or Z) decaying into light quarks ($q'\bar{q}'$).

In figure 8, it starts with a lepton veto because the specific analysis targets fully hadronic ($VV \rightarrow qqqq$) and semi-hadronic/invisible ($ZV \rightarrow \nu\nu qq$) processes, genuine signal events should not contain visible leptons. The first branch is to identify if it exists invisible neutrinos. The DNN tagger in the figure for R-MET is to identify vector bosons from forward jets. After all these event selections are completed, they are forward fed into different structure of DNN model to output the significance.

Variable	Selection requirements
Lepton veto	$e, \mu p_T > 10 \text{ GeV}, \eta < 2.4$ $\tau_h p_T > 18 \text{ GeV}, \eta < 2.3$
b quark veto	Additional with $p_T > 20 \text{ GeV}$
p_T^{miss}	B-B, B-Btag: $< 200 \text{ GeV}$ B-MET, R-MET: $> 200 \text{ GeV}$
Diboson	Object selections in Table 5.17 [9] + “B-B”: $p_T^{\text{leading } AK8} > 400 \text{ GeV}$ B-B, B-Btag: $m_{VV} > 500 \text{ GeV}$ B-MET, R-MET: $m_T > 500, 250 \text{ GeV}$
Forward jets	$N_{\text{jets}}(p_T \geq 50 \text{ GeV}, \eta < 4.7) \geq 2$ $m_{jj} > 500 \text{ GeV}$ $ \Delta\eta_{jj} > 2.5$ $\eta^{j1}\eta^{j2} < 0$
Event p_T imbalance	B-MET, R-MET: $ \vec{p}_T^{\text{tot}} /\sum_i p_T^i < 0.25$

Table 19: Summary of the requirements used to select the VBS signal events.

The full requirements used to select the VBS signal events in all categories are summarized in Table 19. These selections are reported to greatly suppress the non-VBS background.

7/17

18 Particle Transformer

I summarized some worth learning contents from the original paper [10] for Particle Transformer (ParT).

18.1 Input

There are two kind of input mentioned in the paper [10]: particle inputs and interaction inputs.

- Particle Inputs $X_{\text{particle}} \in \mathbb{R}^{N \times C}$: where C is the features per particle. It usually consists of:
 - kinematics: energy, momentum
 - particle identification: charge, particle type
 - trajectory displacement: whether the trajectory deviates from the collision point

- Interaction Inputs $X_{\text{interaction}} \in \mathbb{R}^{N \times N \times C'}$ is the features per paired particles.

- Angular separation Δ :

$$\Delta = \sqrt{(y_a - y_b)^2 + (\phi_a - \phi_b)^2} \quad (36)$$

- Relative transverse momentum k_T :

$$k_T = \min(p_{T,a}, p_{T,b}) \Delta \quad (37)$$

- Momentum sharing variable Z :

$$z = \frac{\min(p_{T,a}, p_{T,b})}{p_{T,a} + p_{T,b}} \quad (38)$$

- Pair invariant mass squared:

$$m^2 = (E_a + E_b)^2 - \|\mathbf{p}_a + \mathbf{p}_b\|^2 \quad (39)$$

18.2 Neural Network

$$f_{\theta}(\text{input}) = \text{output} \quad (40)$$

where θ is the parameters/weights that the machine learns and adjust during training.

18.3 Embedding

To make initial particle features $(p_T, \eta, \phi, E, \text{charge} \dots)$ into a machine learnable vector $x_i^0 \in \mathbb{R}^d$. Therefore, for the whole jet, the embedded input particle features should look like:

$$x^0 = \begin{pmatrix} x_1^0 \\ x_2^0 \\ \vdots \\ x_N^0 \end{pmatrix} \in \mathbb{R}^{N \times d} \quad (41)$$

And for input interaction features:

$$U \in \mathbb{R}^{N \times N \times d'} \quad (42)$$

18.4 Attention

To understand particle i , what other particles we should also be pay attention to? To figure out this, we define the weight α_{ij} , such that the particle embedding becomes:

$$x_i^{\text{new}} = \sum_j \alpha_{ij} V_j \quad (43)$$

where V_j is the information particle j provides.

The transformer than transfer the particle embedding x_i into three kinds of vectors:

- Query projection:

$$Q_i = W_Q x_i \quad (44)$$

“what am I finding? ”

- Key projection:

$$K_i = W_K x_i \quad (45)$$

“what label do I have, can I match with you? ”

- Value projection

$$V_i = W_V x_i \quad (46)$$

“If you choose me, what can I provide? ”

The value for Query-Key dot product is proportional to the importance of particle j to understand particle i .

$$QK^T \quad (47)$$

18.5 Model Structure

- Particle inputs go through the embedding layer to transfer the original particle features into vector x_i^0 for event i .
- Interaction inputs go through another embedding layer to transfer into vector U .
- x^0 go through L particle attention blocks and becomes x^L , while U_{ij} are the interaction bias.
 - Three linear input in the beginning of particle attention block, outputting Query $Q = xW_Q$, Key $K = xW_K$, and Value $V = xW_V$.
 - The MatMul layer acts as QK^T : the particle pair compatibility.
 - The scale layer divides the value by $\sqrt{d_k}$ to prevent scores from saturation.
 - Mask layer tells the model the zero-padding items are not real particle information.
 - LN (linear norm) layer stabilizes the value for different features.
 - $\oplus$ (residual connection): $x' = x + F(x)$ to prevent raw information from easily vanishing within deep networks.
- x_L concatenate with class token (CL) x_{class} , where CL acts as the summarizer for the all the input features.
- Concatenated vector go through two class attention blocks.

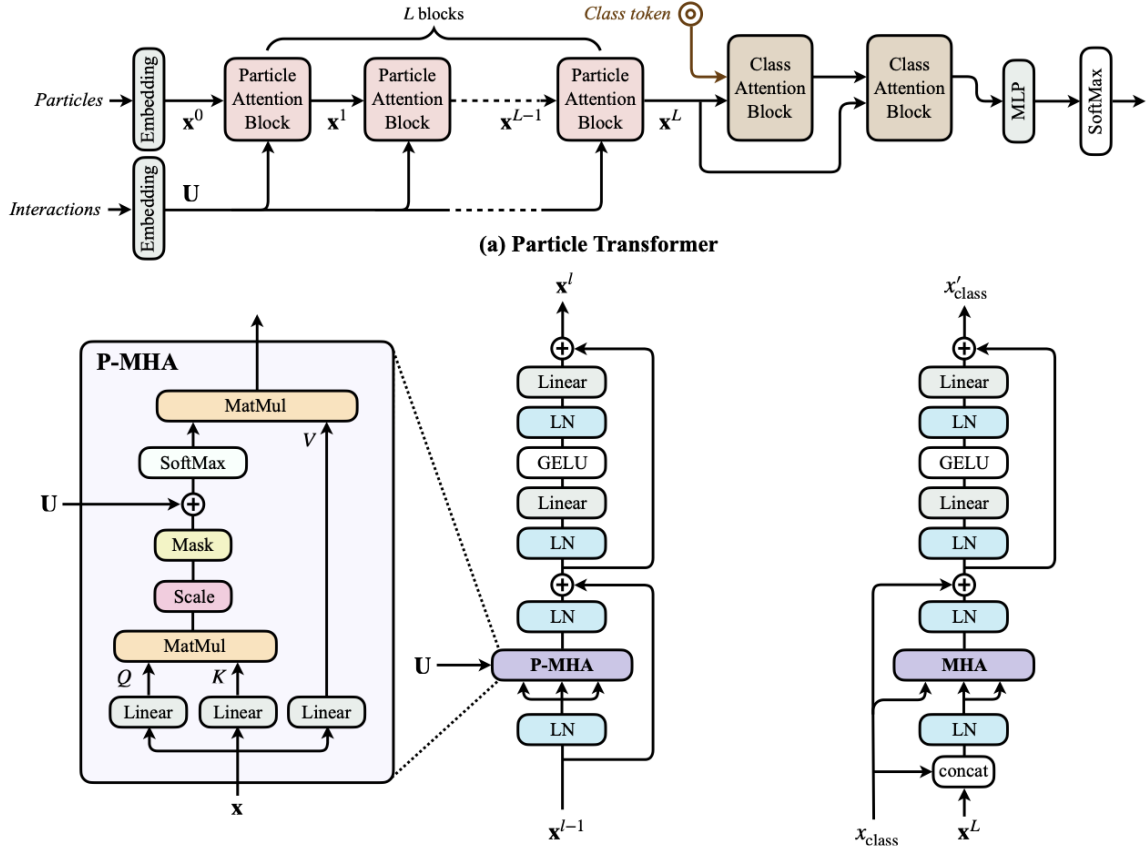


Figure 9: The original model structure in the paper [10]. (a) Overview of the structure of Particle Transformer. (b) Details workflow for particle attention block. (c) Details for class attention block.

- MLP (multi-layer perceptron, linear – GELU –linear): GELU acts as the activation function letting the model to have non-linear patterns.
- Final SoftMax outputs the final score for the probability of different kinds of event type it might be.

$$\text{Attention} = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} + U \right) V_{\text{value}} \quad (48)$$

In conclusion, attention blocks concentrate on how different particles exchange information. While class attention blocks only care about how the class token connects with x_{class} and x_L . And at the end, outputting the final class token x_{class} .

19 Delphes Outputs

19.1 Collider Detector

Inside the collider: collision point – tracker – ECAL – HCAL – muon system

1. Tracker: Charged particles in magnetic field leaving a series of hits and trajectory. Tracker record them and reconstruct them and use particle-flow algorithm to transfer them into particle candidates.
2. Calorimeter: Particles interact with CAL, exciting secondary particles to shower, detectors find record them and work backwards to figure out the energy of the original particle.
 - Electromagnetic Calorimeter (ECAL): Uncharged particles, such as photons, have no trajectories recorded in the tracker. Particles who have electromagnetic interactions will be detected by ECAL. Electrons, Positrons, and Photons are the main objects. The shape of electromagnetic shower is usually more concentrated and short, also having better resolution.
 - Hadronic Calorimeter (HCAL): Uncharged particles, such as neutral hadrons, have no trajectory in tracker. Particles who have strong interactions can be recorded by HCAL. These particles have strong interaction with the nucleus of the material in HCAL, which then excites hadronic shower. Hadronic showers usually have irregular width and depth, while the resolution is worse compare to electromagnetic shower.

19.2 EFlow Constituents

- EFlowTrack: detected by a tracker.
- EFlowPhoton: detected by a ECAL.
- EFlowNeutralHadron: detected by a HCAL.

19.3 Detector Level

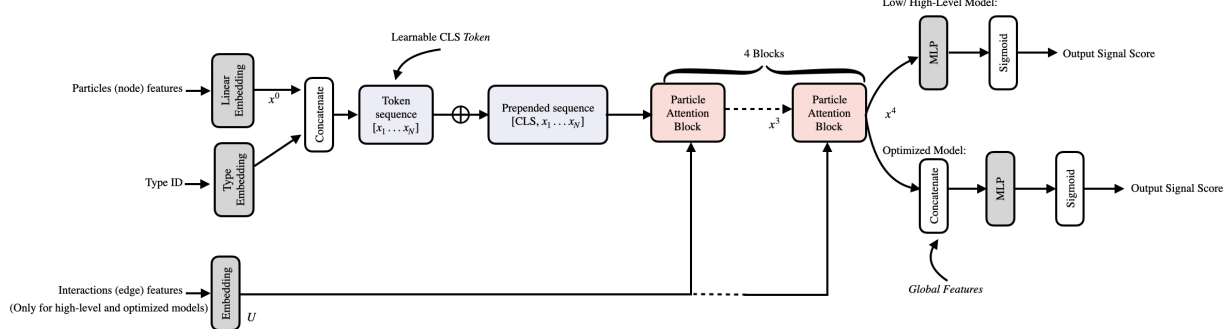
Delphes ROOT file doesn't just store the final layer; it saves multiple layers within the same event.

- Event
 - Low/detector-like collections

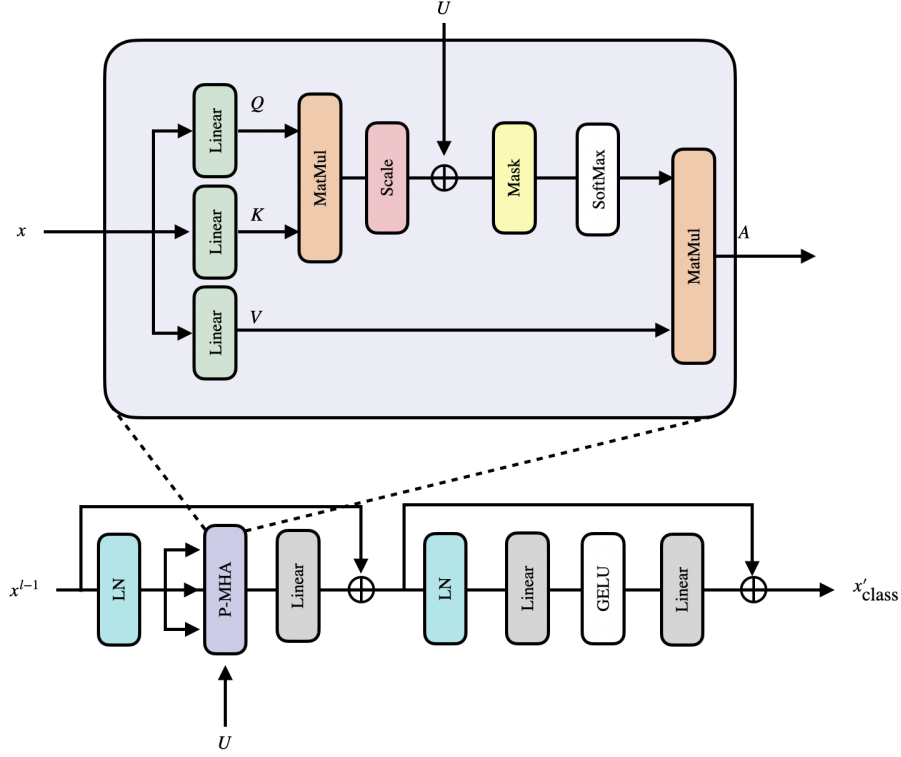
- * Track
 - * Tower
 - * EFlowTrack
 - * EFlowPhoton
 - * EFlowNeutralHadron
- Reconstructed physics objects
 - * Electron
 - * Muon
 - * Jet
- event-level quantity
 - * MET

20 Our Model for ParT

20.1 Model Structure



(a) Overall architecture of the modified Particle Transformer.



(b) Structure of the particle attention block and the incorporation of the pairwise interaction embedding U .

Figure 10: Architecture of the modified Particle Transformer used in this analysis: (a) the complete classification model and (b) the internal structure of a particle attention block.

As shown in Figure 10, our current model is an adapted Particle Transformer to classify event-level, instead of single-jet level. Original ParT set every constituents as particle sequence, we set every reconstructed objects or detector-level constituents as token sequence in each VBS event. Every token

represents a object, such as lepton, jet, MET, or EFlow constituents. The model uses self-attention to learn the kinematic relation, summarize them in CLS token, and finally output the binary signal score. Compare with the original ParT paper [10], we remain the **node/object embedding + type embedding + pairwise attention bias + Transformer attention blocks**.

20.2 Input Features

Table 20: Summary of ParT representations, object shapes, and mathematical feature sets across low, high, and optimized modes.

Mode	Input objects	Node shape	Node features	Edge shape	Edge features	Global shape	Global features
low	EFlowTrack EFlowPhoton EFlowNeutralHadron	128×4	$\log_{10}(p_T \text{ or } E_T)$ η, ϕ $\log_{10}(m)$ type embedding	$128 \times 128 \times 1$	None	5	None
high	lep1, lep2 jet1, jet2 MET	5×4	$\log_{10}(p_T), \eta$ $\Delta\phi$ relative to leading ℓ $\log_{10}(m)$ type: ℓ / jet / MET	$5 \times 5 \times 3$	$\Delta\eta$ $\Delta\phi$ ΔR	5	None
opt	lep1, lep2 jet1, jet2 MET EFlow const.	53×5	$\log_{10}(E_T), \eta$ $\Delta\phi$ relative to leading ℓ $\log_{10}(m), \log_{10}(E)$ type embedding	$53 \times 53 \times 6$	$\Delta\eta, \Delta\phi, \Delta R$ $\log_{10}(m_{\text{pair}})$ $\log_{10}(p_{T,\text{pair}})$ $\log_{10}(m_{T,\text{pair}})$	5	$\log_{10}(m_{jj})$ $\Delta\eta_{jj}$ $Z_{\ell_1}^*, Z_{\ell_2}^*$ $\log_{10}(\text{MET})$

- $\log_{10}(p_T \text{ or } E_T)$: the input value depends on what kind of data delphes output. For example, Track and EFlowTrack use p_T since they come from charged particle trajectory. Photon, NeutralHadron, and Tower use E_T since calorimeter-like objects are usually energy deposit and photon doesn't have mass.
- $\Delta\phi$ relative to leading ℓ : Low model uses raw ϕ since low model is the constituent-level ParT, we show the detector-level EFlow constituents to the model. But for high-model, we use $\Delta\phi$ relative to leading ℓ . Because inside a collider, events rotation around the beam axes shouldn't affect the physics classification. Therefore, the absolute ϕ value itself isn't an important information, it is more important to know the relative angle to each other.
- $\log_{10}(m)$: Objects without meaningful mass, such as MET, EFlowPhoton, EFlowNeutralHadron, are set to zero and use: $\log_{10}(\max(\text{mass}, 1\text{e-}3))$.
- EFlow constituents selection rules: There are several EFlow constituents for each event. We merge the three types of EFlow objects into a single list for each event, sort the list by p_T or E_T in descending order, select the top 128 objects, and apply zero-padding if there are fewer than 128 objects.

A short summary of comparison between our current model and the original model:

Table 21: Comparison between the original Particle Transformer architecture and the modified ParT v4 model used in this work.

Component	Original ParT paper	Our current model
Task	Jet tagging	VBS event classification
Input objects	Jet constituents	High-level physics objects or energy-flow constituents
Class token	Particle-attention blocks $\rightarrow$ token $\rightarrow$ class-attention blocks	CLS token as the initial input
Class-attention blocks	After the particle-attention blocks	No separate class-attention block is used
Output	Multi-class SoftMax classifier	Binary sigmoid classifier
Edge features	Pairwise particle-interaction features are embedded and added to the attention logits	High-level and optimized modes: explicit edge features; Low-level mode: without edge features
Global features	No event-level global features used	The optimized model: concatenates global features with the final CLS-token

As shown in Figure 10, there are three kinds of model trained for different purpose:

- Low-Level: Let the model directly examine constituents that closely resemble the detector output to see if it can learn useful event structures on its own. Input Objects are **EFlowTrack**, **EFlowPhoton**, **EFlowNeutralHadron**. These are particle-flow level constituents, closer to the detector reconstructed particles, but still not the initial detector hits.
- High-Level: To test whether a Transformer can outperform a dense neural network when using only traditional physics objects? Input Objects are **lep1**, **lep2**, **jet1**, **jet2**, **MET**. These are the analysis-level objects, reconstructed by Delphes according to tracker, calorimeter, muon system, jet cluster. Interaction features are incorporated to give more information to the model to improve performance.
- Optimized: Combining two types of information; On one hand, preserving the clear physical meaning of high-level objects; on the other, incorporating EFlow constituents to allow the model to observe finer-grained event activity. Input objects are **lep1**, **lep2**, **jet1**, **jet2**, **MET + leading EFlow constituents**. In addition, global features are incorporated; thus, the optimized model is a physics-informed Particle Transformer.

20.3 Hyper-parameters

Table 22: Model Hyperparameters and Training Configuration

Hyperparameter	Value
optimizer	AdamW
learning rate	5e-4
weight decay	1e-4
batch size	256
max epochs	100
patience	8
embedding dim	64
attention heads	4
Transformer layers	4
dropout	0.2
loss	weighted BCELoss
gradient clipping	5.0
folds	5, but current results are fold0 only

20.4 Results

I summarized the performance, including the AUC value, the expected discovery significance Z_0 , the 95% CL upper limit, the training time, and the best epoch for three kinds of modes.

Table 23: Performance, upper-limit results, and training information for the low-, high-, and optimized-level ParT representations.

Task	Mode	AUC	Z_0	Upper limits μ_{95}, σ_{95} [fb]	Training time	Best epoch
WW	low	0.8757	24.5697	0.0868, 0.0741	52.0 min	24
	high	0.7923	20.4625	0.0885, 0.0756	7.3 min	5
	opt	0.9441	29.3286	0.0845, 0.0722	52.1 min	46
LL	low	0.8308	2.1456	0.9592, 0.4249	112.0 min	30
	high	0.8455	2.2392	0.9164, 0.4060	15.3 min	7
	opt	0.8429	2.2227	0.9223, 0.4086	28.5 min	4
LX	low	0.7064	6.0093	0.3011, 0.5691	58.2 min	11
	high	0.7586	6.9991	0.2503, 0.4731	34.1 min	26
	opt	0.7590	6.9767	0.2510, 0.4743	65.5 min	26

In Figure 23, we can see that for *WW* models, optimized model has the best performance while low-level model also performed better than previous dense-NN model. Meaning that giving out global features did help the model to classify signal from background pretty well. For *LL* model, things became different. High-level model has the best performance among the other two. Noticed that *LX* model performs extremely worse, this might be something wrong in the code that we can find out also in Figure 11 for the unusual loss functions and ROC curves.

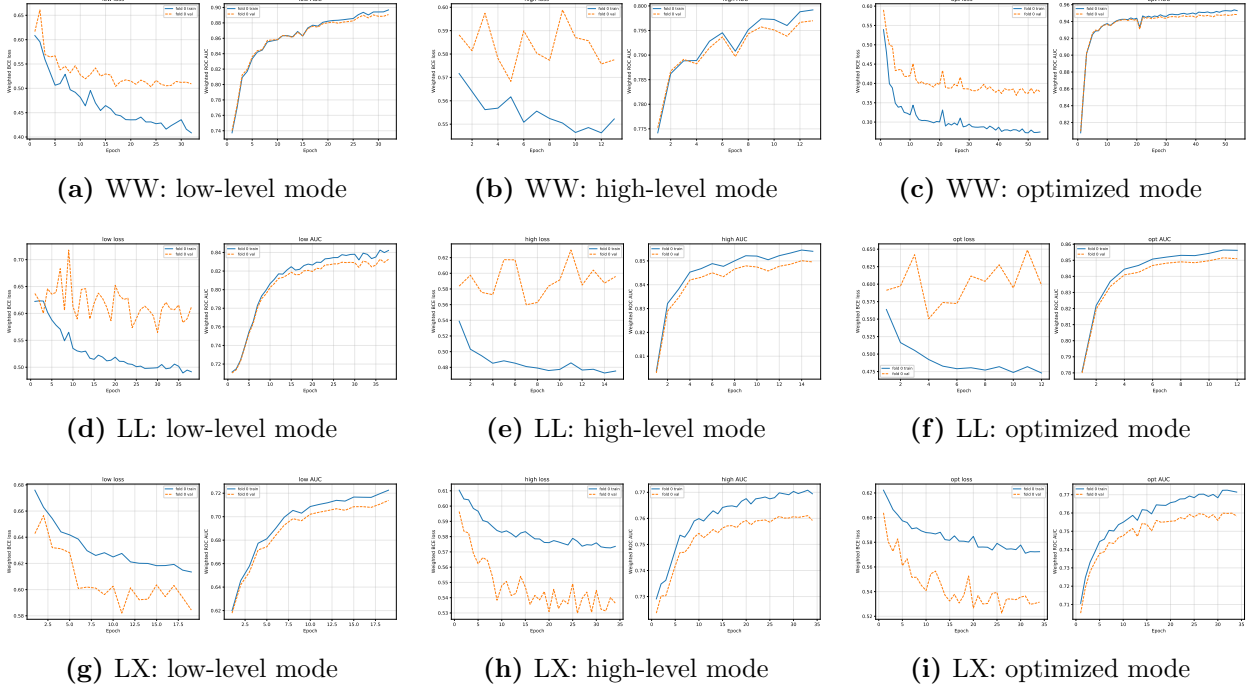


Figure 11: Training curves for the WW, LL, and LX classification tasks using the low-level, high-level, and optimized ParT representations. Each panel shows the weighted binary cross-entropy loss and weighted ROC AUC for the training and validation samples.

In Figure 11, the loss functions for high-level models seems to oscillate too violent. Meaning that maybe for epochs are needed. While I am only using 8 patience for the training. For *LX* models, there might be some code error or training problem we need to figure out since the output training curves seem to be weird and abnormal.

See Figure 12 to visualize the classification ability for each model.

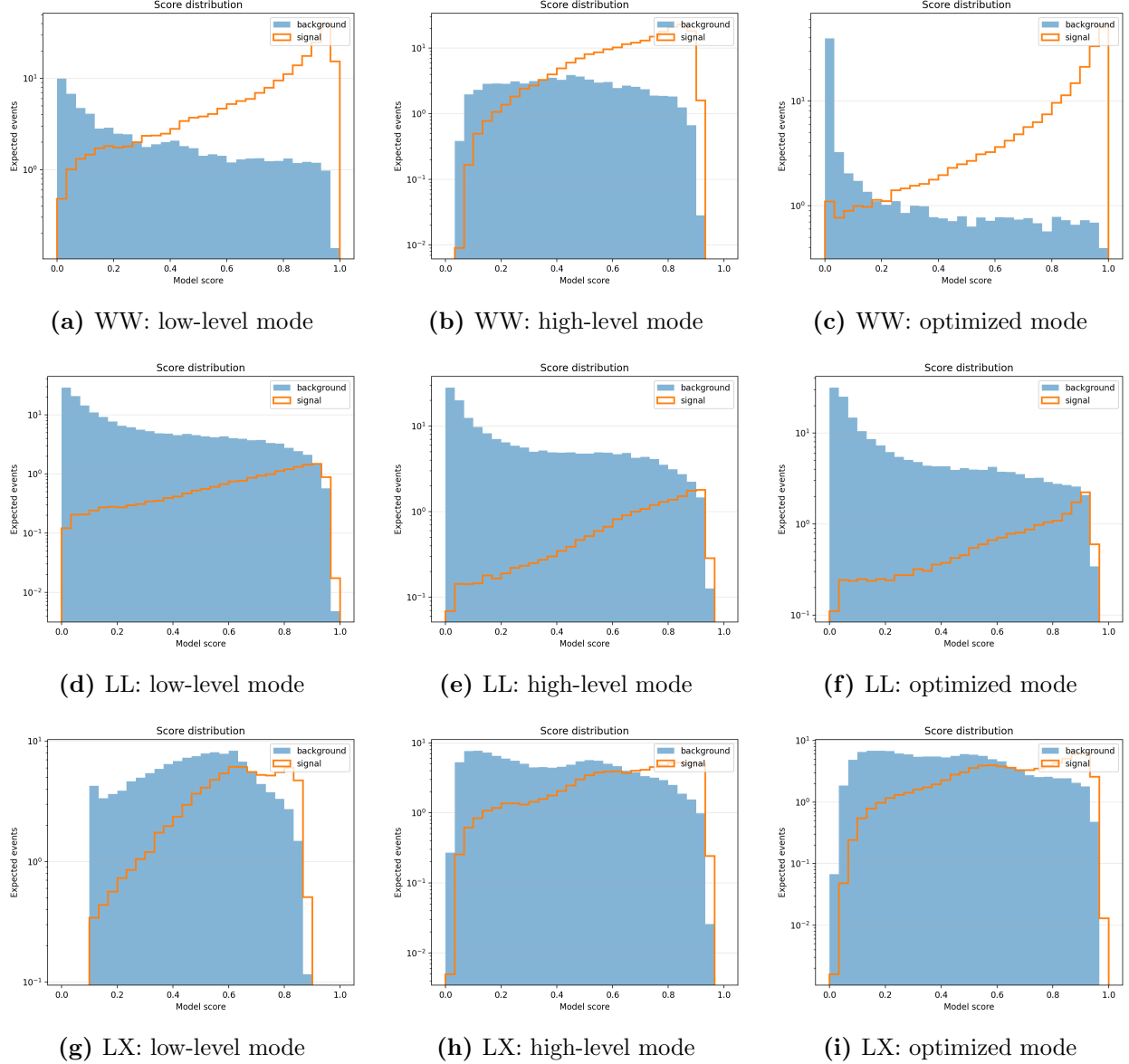


Figure 12: Model-score distributions for the WW, LL, and LX classification tasks using the low-level, high-level, and optimized ParT representations. The filled histograms show the expected background distributions, while the outlined histograms show the corresponding signal distributions for fold 0.

Figure 12 shows that signal events are generally shifted toward larger model scores, while background events preferentially populate lower scores. The separation is most pronounced for the WW task, particularly in the optimized representation, whereas the LL and LX tasks exhibit greater signal–background overlap.

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